Eidgenössische Technische Hochschule Zürich

Funktionentheorie

Autumn Semester 2026

Lecture Notes

Contents

1 Introdcution **1**
1.1 Motivation . . . 1
1.2 Review of Complex Numbers . . . 1
1.3 Notations from Analysis . . . 2
1.4 Holomorphic Functions . . . 3
1.5 Cauchy-Riemann equations . . . 4
1.6 Power Series . . . 5
1.7 Complex Line Integrals . . . 6

2 Cauchy's Theorem on a disc **8**
2.1 Sequences of holomorphic functions . . . 14

3 Meromorphic Functions and the Logarithm **17**

References

[1] Imamoglu, Özlem: Vorlesungsnotizen Funktionentheorie. ETH Zürich. 2026

Funktionentheorie

Fynn Krebser–fkrebser@student.ethz.ch

Autumn Semester 2026

1 Introdcution

Lec 1

1.1 Motivation

Consider the following example. On $\mathbb{R}$ it is not too difficult to construct a function f that is differentiable n times but not differentiable $n+1$ times. For example,

$$f(x) = \begin{cases} x^2 \sin(\frac{1}{x^2}) & x \neq 0, \\ 0 & x = 0 \end{cases}.$$

This function is differentiable but the derivative is not continuous. Integrating, we find a function that is differentiable twice but not three times. In fact, we can construct a function that is differentiable n times but not $n+1$ times for any $n \in \mathbb{N}$ using this method.

In contrast, we'll see that if $f : \mathbb{C} \to \mathbb{C}$ is differentiable once, then it is differentiable infinitely many times.

Another example is that for real numbers, there are functions $f : \mathbb{R} \to \mathbb{R}$ which are infinitely differentiable but not analytic. I.e. the Taylor series at some point x_0 exists, but does not converge to the function in any neighborhood of x_0. For example,

$$x \longmapsto (x) = \begin{cases} e^{-\frac{1}{x^2}} & x \neq 0, \\ 0 & x = 0 \end{cases},$$

where all derivatives are zero so the Taylor series is identically zero.

In contrast, we'll see that a differentiable function $f : \mathbb{C} \to \mathbb{C}$ has a power series representation. So differentiability is equivalent to analyticity.

A further comparison is that there are many examples of functions $f : \mathbb{R} \to \mathbb{R}$ which are C^∞ and bounded, for example

$$f(x) = \sin(x).$$

In contrast, we'll see that if $f : \mathbb{C} \to \mathbb{C}$ is bounded and differentiable, then it is constant.

The robustness of this theory can be used for applications which seem to have nothing to do with complex analysis. Probably the most used examples are evaluating real integrals.

$$\int_0^\infty \cos(t^2)\mathrm{d}t = \int_0^\infty \sin(t^2)\mathrm{d}t = \frac{\sqrt{2\pi}}{4}.$$

Another famous example is the prime number theorem, which states that the number of primes less than x is asymptotic to $\frac{x}{\log(x)}$.

Another classic is to consider a non-zero polynomial $f(x) \in \mathbb{C}[x]$. Then the Fundamental Theorem of Algebra states that f has a zero in $\mathbb{C}$.

Lastly, consider

$$r_4(n) = \#\{(x,y,z,w) \in \mathbb{Z}^4 : x^2+y^2+z^2+w^2 = n\}.$$

Lagrange proved that every positive integer has such a representation, namely $r_4(n) \geq 1$. Jacobi later proved that

$$r_4(n) = 8 \sum_{d|n;4\nmid d} d.$$

1.2 Review of Complex Numbers

Recall that the complex numbers are given as

$$\mathbb{C} = \{x + iy : x, y \in \mathbb{R}\},$$

where $i^2 = -1$. We can embed $\mathbb{R}$ into $\mathbb{C}$ via $x \mapsto x + i0$. We also defined

Definition 1.1: Real and Imaginary Parts

For $z = x+iy \in \mathbb{C}$, we define the REAL PART $\Re(z) = x$ and the IMAGINARY PART $\Im(z) = y$.

Furthermore, the COMPLEX CONJUGATE is defined as

$$\overline{z} = x - iy.$$

From this it follows that

Lemma 1.2:

For $z = x + iy \in \mathbb{C}$, we have

$$z \in \mathbb{R} \Leftrightarrow z = \overline{z} \qquad z \in i\mathbb{R} \Leftrightarrow z = -\overline{z}.$$

$\mathbb{C}$ can also be represented by pairs of real numbers in $\mathbb{R}^2$. For example $z = (x,y), w = (u,v) \in \mathbb{R}^2$ where

$$z = w \Leftrightarrow x = u, y = v.$$

Addition can be defined in both notations as

$$z + w = (x+u, y+v) \quad z + w = (x+w) + i(y+v).$$

Multiplication was a bit more cumbersome. Using the definition of i, we have

$$\begin{aligned} z \cdot w &= (x+iy)(u+iv) \\ &= xu + i^2yv + i(xv+yu) \\ &= (xu - yv) + i(xv+yu). \end{aligned}$$

So in the pair representation, multiplication is given by

$$(x,y) \cdot (u,v) = (xu - yv, xv + yu).$$

Note that in the pair representation,

$$i \leftrightarrow (0,1) \quad \text{and} \quad 1 \leftrightarrow (1,0).$$

The good thing about the pair notation, $(\mathbb{R}^2, +, \cdot)$ satisfies

1. $(\mathbb{R}^2, +)$ is an abelian group with identity $(0,0)$
2. $(\mathbb{R}^2 \setminus \{(0,0)\}, \cdot)$ is an abelian group with identity $(1,0)$
3. The distributive law holds: $z_1(z_2+z_3) = z_1z_2 + z_1z_3$

Together, these properties make $(\mathbb{R}^2, +, \cdot)$ a field. So the complex numbers with the above addition and multiplication form a field.

The additive inverse of $z = (x,y)$ is given by $-z = (-x,-y)$. The multiplicative inverse of $0 \neq z = (x,y)$ is given by

$$z^{-1} = \frac{\overline{z}}{|z|^2} = \frac{(x,-y)}{x^2+y^2} = \left(\frac{x}{x^2+y^2}, -\frac{y}{x^2+y^2}\right).$$

Here, $|z| \in \mathbb{R}$ is the MODULUS or NORM of z, defined as

$$|z| = \sqrt{x^2+y^2}.$$

We can also consider the POLAR COORDINATES of z. Any point on the plane $\mathbb{R}^2$ can be represented by a distance r from the origin and an angle θ from the positive x-axis. Here,

$$z = r(\cos(\theta) + i\sin(\theta)) = re^{i\theta}.$$

Beware that the representation is not unique, since θ can be replaced by $\theta + 2\pi k$ for any $k \in \mathbb{Z}$, unless we restrict θ to a certain interval. The standard is to take $\theta \in (-\pi, \pi]$. θ is called the ARGUMENT of z, denoted $\arg(z)$, and r is the modulus $|z|$. For the argument in $(-\pi, \pi]$, we write $\mathrm{Arg}(z)$ and call it PRINCIPAL ARGUMENT.

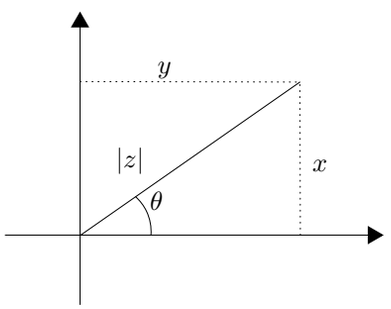


Figure 1: Polar coordinates of a complex number z

Lec 2

The principal argument of $z = x + iy \neq 0$ we have

$$\mathrm{Arg}(z) = \begin{cases} \arcsin(\frac{y}{|z|}) & x \geq 0 \\ \pi - \arcsin(\frac{y}{|z|}) & x < 0, y \geq 0 \\ -\pi - \arcsin(\frac{y}{|z|}) & x < 0, y < 0 \end{cases}.$$

Here, for $t \in [-1,1]$, $\arcsin(t)$ is the unique number u in $[-\frac{\pi}{2}, \frac{\pi}{2}]$ such that $\sin(u) = t$.

Notice, that

$$\arg(z^-1) = -\arg(z) \quad \text{and} \quad \arg(zw) = \arg(z) + \arg(w).$$

But, it is not always the case if we use the principal argument instead. For example

$$\mathrm{Arg}(-\frac{1}{2}) = \pi \neq -\mathrm{Arg}(-2) = -\pi.$$

Lemma 1.3:

Let $z, w \in \mathbb{C}$. Then

- $|z| = 0 \Leftrightarrow z = 0$
- $||z| - |w|| \leq |z+w| \leq |z| + |w|$
- $|zw| = |z||w|$
- $|\overline{z}| = |z|$
- $|\Re(z)| \leq |z|$ and $|\Im(z)| \leq |z|$

If we have two numbers $z_1 = r_1e^{i\theta_1}$ and $z_2 = r_2e^{i\theta_2}$ in polar form, then

$$z_1z_2 = r_1r_2e^{i(\theta_1+\theta_2)}.$$

$\mathbb{C}$ also has a representation in terms of 2×2 matrices. For $z = a + ib$, let

$$Z = \begin{pmatrix} a & -b \\ b & a \end{pmatrix}.$$

If $w = c + id$, then

$$ZW = \begin{pmatrix} ac-bd & -(bc+ad) \\ bc+ad & ac-bd \end{pmatrix}.$$

So we can do the multiplication of complex numbers using matrix multiplication.

We can also write the matrix representation as

$$Z = \Re(z)\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \Im(z)\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

In polar form, $z = re^{i\theta}$, we have

$$Z = \begin{pmatrix} r & 0 \\ 0 & r \end{pmatrix}\begin{pmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{pmatrix}.$$

Notice that the second matrix lives in $SO(2)$, the group of rotations in $\mathbb{R}^2$.

1.3 Notations from Analysis

We like to recall some definitions from analysis regarding open and closed sets.

Definition 1.4: Discs

The OPEN DISC of radius r centered at z is the set

$$D_r(z) = \{w \in \mathbb{C} : |w - z| < r\}.$$

The CLOSED DISC of radius r centered at z is the set

$$\overline{D}_r(z) = \{w \in \mathbb{C} : |w - z| \leq r\}.$$

The boundary circle is the set

$$C_r(z) = \{w \in \mathbb{C} : |w - z| = r\}.$$

Definition 1.5: Connected and Disconnected Sets

A subset $A \subset \mathbb{C}$ is called DISCONNECTED if there exists two open sets $U, V \subset \mathbb{C}$ such that

$$U \cap V = \varnothing, \quad A \cap U \neq \varnothing, \quad A \cap V \neq \varnothing, \quad A \subset U \cup V.$$

A set is called CONNECTED if it is not disconnected. A connected open subset $U \neq \varnothing$ of $\mathbb{C}$ is called a REGION or a DOMAIN.

1.4 Holomorphic Functions

We now want to define what it means for a complex function to be differentiable.

Definition 1.6: Holomorphic functions

Let $\Omega \subset \mathbb{C}$ be an open and $f : \Omega \to \mathbb{C}$ a function.

f is called DIFFERENTIABLE or HOLOMORPHIC at $z_0 \in \Omega$ if

$$\lim_{z \to z_0, z \neq z_0} \frac{f(z) - f(z_0)}{z - z_0} = \lim_{h \to 0, h \neq 0} \frac{f(z+h) - f(z)}{h}.$$

If f is differentiable $\forall z \in \Omega$, it is holomorphic on Ω. If $\Omega = \mathbb{C}$ and f is holomorphic on $\mathbb{C}$ then f is called ENTIRE.

The value of the limit is denoted by $f'(z_0)$.

Example 1.7:

Consider $f : \mathbb{C} \to \mathbb{C}, z \mapsto z$. Then f is entire. Indeed,

$$\lim_{h \to 0, h \neq 0} \frac{f(z+h) - f(z)}{h} = \lim_{h \to 0, h \neq 0} \frac{z + h - z}{h} = 1.$$

Proposition 1.8: Properties of holomorphic functions

1. Let $\mathcal{H}(\Omega)$ be the set of holomorphic functions on Ω. Then $\mathcal{H}(\Omega)$ is a vector space over $\mathbb{C}$.

Here, $O : z \mapsto 0$ is the 0-element.

2. If $f, g \in \mathcal{H}(\Omega)$ then so is $f \cdot g$. Also, we have

$$(fg)'(z_0) = f'(z_0)g(z_0) + g'(z_0)f(z_0).$$

3. Lastly if $g(z_0) \neq 0$, for any $z_0 \in \Omega$, then f/g is holomorphic at z_0 and

$$\left(\frac{f}{g}\right)'(z_0) = \frac{f'(z_0)g(z_0) - g'(z_0)f(z_0)}{g(z_0)^2}.$$

4. If $f : \Omega \to U$ and $g : U \to \mathbb{C}$ are holomorphic, then $g \circ f$ is holomorphic and

$$(g \circ f)'(z_0) = g'(f(z_0))f'(z_0).$$

Proof. The proof is the same as in real analysis. We will thus only show the proof of the last statement.

Let $z_0 \in \Omega$ and let $f(z_0) = w_0 \in U$. Consider

$$F : \Omega \to \mathbb{C}, G : U \to \mathbb{C}.$$

where

$$F(z) = \begin{cases} \frac{f(z)-f(z_0)}{z-z_0} & z \neq z_0 \\ f'(z_0) & z = z_0 \end{cases}, \quad G(z) = \begin{cases} \frac{g(w)-g(w_0)}{w-w_0} & w \neq w_0 \\ g'(w_0) & w = w_0 \end{cases}.$$

Since f and g are holomorphic, F and G are continuous. Hence, $G \circ f$ is continuous at z_0.[1] Now,

$$\frac{(g \circ f)(z) - (g \circ f)(z_0)}{z - z_0} = \frac{g(f(z)) - g(f(z_0))}{f(z) - f(z_0)}.$$

Using $f(z_0) = w_0$ we can write this as

$$\frac{g(f(z)) - g(w_0)}{f(z) - w_0} \cdot \frac{f(z) - f(z_0)}{z - z_0} = G(f(z))F(z).$$

Note that if $f(z) = w_0$ we get that $F(z) = \frac{w_0 - f(z_0)}{z - z_0} = 0$. Hence, taking the limit $z \to z_0$ and using continuity, we get that

$$\begin{aligned} \lim_{z \to z_0, z \neq z_0} \frac{(g \circ f)(z) - (g \circ f)(z_0)}{z - z_0} &= \lim_{z \to z_0, z \neq z_0} G(f(z))F(z) \\ &= G(f(z_0))F(z_0) \\ &= g'(f(z_0))f'(z_0). \end{aligned}$$

□

Remark

If $f : \Omega \to \mathbb{C}$ is differentiable at $z_0 \in \Omega$, then $\exists$ a complex number $c \in \mathbb{C}$ such that

$$f(z) = f(z_0) + c(z - z_0) + o(|z - z_0|) \quad \text{as } z \to z_0.$$

Here, for c we write $f'(z_0)$.

Example 1.9: Polynomials are entire

Using $f(z) = z$ entire and Proposition 1.8, we get that any polynomial is entire. If $g(z) = z^n$, then

$$g'(z) = nz^{n-1}.$$

Example 1.10: Real and complex differentiability

Let $f(z) = \overline{z}$. Then

$$\frac{f(z_0 + h) - f(z_0)}{h} = \frac{\overline{z_0 + h} - \overline{z_0}}{h} = \frac{\overline{h}}{h}.$$

If we choose $h = t, t \in \mathbb{R}$, then $\frac{\overline{h}}{h} = 1$. If we choose $h = it, t \in \mathbb{R}$, then $\frac{\overline{h}}{h} = -1$. Hence, the limit does not exist and f is not holomorphic.

If we look at f as a function from $\mathbb{R}^2 \to \mathbb{R}^2$, then $\tilde{f}(x, y) = (x, -y)$. Then $\tilde{f}$ is differentiable in the real sense and the Jacobian is given by

$$D\tilde{f}(x, y) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Recall, that a function $f : \mathbb{R}^2 \to \mathbb{R}^2, (x, y) \mapsto (u, v)$ is differentiable at (x_0, y_0) if there exists a linear map $L : \mathbb{R}^2 \to \mathbb{R}^2$ such that

$$\lim_{(x,y) \to (x_0, y_0)} \frac{|f(x, y) - f(x_0, y_0) - L(x - x_0, y - y_0)|}{|(x - x_0, y - y_0)|} = 0.$$

The linear map L is unique and called the DIFFERENTIAL Df. In the standard basis, the linear map L has a matrix representation which is called the JACOBIAN of f. It is given by

$$Jf(x_0, y_0) = \begin{pmatrix} \frac{\partial u}{\partial x}(x_0, y_0) & \frac{\partial u}{\partial y}(x_0, y_0) \\ \frac{\partial v}{\partial x}(x_0, y_0) & \frac{\partial v}{\partial y}(x_0, y_0) \end{pmatrix}.$$

[1] f is differentiable at z_0 so continuous at z_0.

Recall from linear algebra, that we can view $\mathbb{C}$ as a vector space over $\mathbb{C}$ with dimension 1 with basis $\{1\}$. Alternatively we can view $\mathbb{C}$ as a vector space over $\mathbb{R}$ with dimension 2 with basis $\{1, i\}$.

A map $T : \mathbb{C} \to \mathbb{C}$ is $\mathbb{C}$-linear if $Tz = \lambda z$ for some $\lambda \in \mathbb{C}$. It is $\mathbb{R}$-linear if

$$T(x + iy) = xT(1) + yT(i) \quad \forall x, y \in \mathbb{R}.$$

Using $x = \frac{z+\overline{z}}{2}$ and $y = \frac{z-\overline{z}}{2i}$, we can write

$$T(z) = \underbrace{\frac{T(1) - iT(i)}{2}}_{:=\lambda} z + \underbrace{\frac{T(1) + iT(i)}{2}}_{:=\mu} \overline{z}.$$

Since T is linear, the coefficient of $\overline{z}$ must be 0. So every $\mathbb{C}$-linear map is $\mathbb{R}$-linear if we set $\mu = 0$. However an $\mathbb{R}$-linear map is only $\mathbb{C}$-linear if $\mu = 0$, so

$$T(i) = iT(1).$$

If $T(1) = a + ib$ and $T(i) = c + id$, then the condition $T(i) = iT(1)$ is equivalent to

$$a = d \quad \text{and} \quad b = -c.$$

Every $\mathbb{R}$-linear map $\mathbb{R}^2 \to \mathbb{R}^2$ is given by a 2×2 matrix $\begin{pmatrix} a & c \\ b & d \end{pmatrix}$. If this $\mathbb{R}$-linear map is also $\mathbb{C}$-linear, then on the matrix we have $a = d$ and $b = -c$. Hence, the matrix is of the form

$$\begin{pmatrix} a & b \\ -b & a \end{pmatrix}.$$

Lec 3

1.5 Cauchy-Riemann equations

Let $f : \mathbb{C} \to \mathbb{C}$ be a holomorphic function at $z = z_0$. If we write $f(x+iy) = u(x, y) + iv(x, y)$. If we view f as a map from $\mathbb{R}^2 \to \mathbb{R}^2$, if f is differentiable at z_0, then

$$f'(z_0) = \lim_{h \to 0} \frac{f(z_0 + h) - f(z_0)}{h} = \lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0}.$$

The limit exists and is independent of the way we approach z_0. In particular, we can approach z_0 along the line $z = x + iy_0$ and let $x \to x_0$. Thus,

$$\begin{aligned} f'(z_0) &= \lim_{x \to x_0} \frac{f(x + iy_0) - f(x_0 + iy_0)}{x - x_0} \\ &= \lim_{x \to x_0} \frac{u(x - x_0) - u(x_0, y_0)}{x - x_0} + i \lim_{x \to x_0} \frac{v(x, y_0) - v(x_0, y_0)}{x - x_0} \\ &= u_x(z_0) + iv_x(z_0). \end{aligned}$$

We can also approach z_0 along the line $z = x_0 + iy$ and let $y \to y_0$. Doing the same calculation, we find

$$f'(z_0) = v_y(z_0) - iu_y(z_0).$$

Combining the equations, we find the Cauchy-Riemann equations

Theorem 1.11: Cauchy-Riemann equations

For a function $f : \mathbb{C} \to \mathbb{C}$, $f(z) = u(z) + v(z)$, if f is holomorphic at z_0, then

$$u_x(z_0) = v_y(z_0), \quad v_x(z_0) = -u_y(z_0).$$

Definition 1.12:

We define the operator

$$\frac{\partial}{\partial z} := \frac{1}{2}\left(\frac{\partial}{\partial x} - i\frac{\partial}{\partial y}\right)$$

and

$$\frac{\partial}{\partial \overline{z}} := \frac{1}{2}\left(\frac{\partial}{\partial x} + i\frac{\partial}{\partial y}\right)$$

Proposition 1.13:

If f is holomorphic at z_0, $f(z) = u + iv$, then

$$\frac{\partial f}{\partial \overline{z}} = 0,$$

and

$$f'(z_0) = \frac{\partial}{\partial z} f(z_0) = 2\frac{\partial u}{\partial z}(z_0).$$

If we write $f(z) = \tilde{f}(x, y)$ as a function from $\mathbb{R}^2$ to $\mathbb{R}^2$, then $\tilde{f}$ is differentiable with Jacobian

$$J_{\tilde{f}}(x_0, y_0) = \begin{pmatrix} u_x & u_y \\ v_x & v_y \end{pmatrix} = \begin{pmatrix} u_x & u_y \\ -u_y & u_x \end{pmatrix}.$$

Furthermore, $\det J_f = |f'(z_0)|^2 = u_x^2 + u_y^2 = u_x^2 + v_x^2$

Proof. Per definition

$$\begin{aligned} \frac{\partial f}{\partial \overline{z}} &= \frac{1}{z}\left(\frac{\partial}{\partial x} + i\frac{\partial}{\partial y}\right)(u + iv) \\ &= \frac{1}{2}(u_x + iv_x + i(u_y + iv_y)) \\ &= \frac{1}{2}(u_x - v_y + i(v_x + u_y)) = 0, \end{aligned}$$

where we used the Cauchy Riemann equations in the last step.

Using the equations we derived for the CR-Equations we get

$$f'(z_0) = \frac{1}{2}(u_x + iv_x + v_y - iu_y) = \frac{1}{2}(f_x(z_0) + -if_y(z_0)).$$

But this is exactly $\frac{\partial f}{\partial z} f(x_0)$

The CR equation give

$$f'(z_0) = \frac{\partial}{\partial z} f(z_0) = u_x + iv_x = u_x - iu_y.$$

Reformulating, we find the result.

If $z_0 = x_0 + iy_0$ and $h = h_1 + h_2 i$, then since f is holomorphic at z_0,

$$f(z_0 + h) = f(z_0) + f'(z_0)h + h\varepsilon(h),$$

where $\lim_{h \to 0} \varepsilon(h) = 0$.

If $f'(z_0) = a + ib$, where $f' = u_x + iv_x = v_y - iu_y$, then

$$\begin{aligned} f'(z_0)h &= (a + ib)(h_1 + h_2 i) \\ &= (ah_1 - bh_2) + i(bh_1 + ah_2). \end{aligned}$$

If we write $\tilde{f}(x, y) = f(z)$ and $H = (h_1, h_2)$, then

$$\left[\tilde{f}((x_0, y_0) + (h_1, h_2)) - \tilde{f}(x_0, y_0) - \begin{pmatrix} a & -b \\ b & a \end{pmatrix}\begin{pmatrix} h_1 \\ h_2 \end{pmatrix}\right] / |H|,$$

goes to zero.

This says, that $\tilde{f} : \mathbb{R}^2 \to \mathbb{R}^2$ is differentiable with

$$J_{\tilde{f}}(x_0, y_0) = \begin{pmatrix} a & -b \\ b & a \end{pmatrix} = \begin{pmatrix} u_x & -v_x \\ v_x & u_x \end{pmatrix}.$$

Hence, $\det J_{\tilde{f}} = a^2 + b^2 = |f'(z_0)|^2 = u_x^2 + v_x^2 = u_x^2 + u_y^2$.

□

Lec 4

Note, that the first equation,

$$\frac{\partial f}{\partial \overline{z}} = 0,$$

is a compact way to write the two Cauchy-Riemann equations.

Theorem 1.14:

Suppose $f = u + iv$ is a function $f : \Omega \to \mathbb{C}$. If u and v are continuously differentiable and satisfy the Cauchy-Riemann equations on Ω, then f is holomorphic.

Note, that it would only be nesessary to have that u, v are totally differentiable.

Proof. Let $z_0 = (x_0, y_0) \in \Omega$ and $h = (h_1, h_2) \in \mathbb{C}$. Since u, v are continuously differentiable, we have that u_x, u_y, v_x, v_y exist and are continuous. From real analysis, we know that this implies that u and v are totally differentiable.

Hence,

$$u(z_0 + h) - u(z_0) = \partial_x u(z_0)h_1 + \partial_y u(z_0)h_2 + |h|\epsilon_1(h),$$

where $\epsilon_1(h) \to 0$ as $h \to 0$. Similarly,

$$v(z_0 + h) - v(z_0) = \partial_x v(z_0)h_1 + \partial_y v(z_0)h_2 + |h|\epsilon_2(h).$$

Combining these two equations, we have, letting $\epsilon(h) = \epsilon_1(h) + i\epsilon_2(h)$,

$$\begin{aligned} f(z_0 + h) - f(z_0) &= (u + iv)(z_0 + h) - (u + iv)(z_0) \\ &= (\partial_x u + i\partial_x v)h_1 + (\partial_y u + i\partial_y v)h_2 + |h|\epsilon(h) \\ &= (\partial_x u - i\partial_y u)h_1 + (\partial_y u + i\partial_x u)h_2 + |h|\epsilon(h) \\ &= (\partial_x u - i\partial_y u)(h_1 + ih_2) + |h|\epsilon(h). \end{aligned}$$

Since $\epsilon(h) \to 0$ as $h \to 0$, we have that

$$\lim_{h\to 0} \frac{f(z_0 + h) - f(z_0)}{h} = \partial_x u - i\partial_y u.$$

This implies, that f is differentiable at z_0 and its derivative is given by

$$\partial_x u - i\partial_y u = 2\frac{\partial}{\partial z} u.$$

□

Example 1.15: Application of Cauchy-Riemann

Let $f(z) = \underbrace{x^2 + y^2}_{u} + i\,\underbrace{2xy}_{v}$. When is f holomorphic?

We have

$$u_x = 2x, \quad u_y = 2y, \quad v_x = 2y, \quad v_y = 2x.$$

So u, v are continuously differentiable. They satsify the Cauchy-Riemann equations if and only if $2x = 2x$ and $2y = -2y$, which is true if and only if $y = 0$. Hence, f is holomorphic on the real line.

We've seen that for f holomorphic, $J_f = \begin{pmatrix} u_x & -v_x \\ v_x & u_x \end{pmatrix}$. Recall from linear algebra, that a linear map of the form $\begin{pmatrix} a & -b \\ b & a \end{pmatrix}$ is a map which preserves angles, as it is a composition of a rotation and a scaling. This follows, since a linear map $z \mapsto (z)(a + ib)$, when written in polar coordinates, is a rotation by $\arg(a + ib)$ and a scaling by $|a + ib|$.

Hence, a holomorphic map, locally preserves angles.

1.6 Power Series

A power series is a series of the form

$$\sum_{n=0}^{\infty} a_n z^n.$$

Theorem 1.16:

Let $\sum a_n z^n$ be a power series. Then $\exists R \in \mathbb{R}_{\geq 0} \cup \{\infty\}$ such that

1. If $|z| < R$, then the series converges absolutely.
2. If $|z| > R$, then the series diverges.

Moreover, with the convention that $\frac{1}{0} = \infty$ and $\frac{1}{\infty} = 0$, R is given by

$$\frac{1}{R} = \limsup_{n\to\infty} |a_n|^{1/n}.$$

R is called the RADIUS OF CONVERGENCE of the power series. $D_R(0)$ is called the DISK OF CONVERGENCE.

Proof. The proof of this is the same as in real analysis, and is left as an exercise. □

Example 1.17: Exponential Function

The complex exponential function is defined by the power series

$$\exp(z) := \sum_{n=0}^{\infty} \frac{z^n}{n!}.$$

It converges absolutely for all $z \in \mathbb{C}$, since

$$\limsup_{n\to\infty} |1/n!|^{1/n} = 0.$$

We will see, that e^z in particular and power series in general give holomorphic functions in their disk of convergence.

Theorem 1.18:

The power series $\sum_{n=0}^{\infty} a_n z^n$ defines a holomorphic function on $D_R(0)$. It's derivative is given by

$$\left(\sum_{n=0}^{\infty} a_n z^n\right)' = \sum_{n=1}^{\infty} n a_n z^{n-1}.$$

Moreover, f' has the same disk of convergence as f.

Proof. Let R be the radius of convergence of f. Since $\lim_{n\to\infty} n^{\frac{1}{n}} = 1$, we have that $\limsup_{n\to\infty} |na_n|^{1/n} = \limsup_{n\to\infty} |a_n|^{1/n} = \frac{1}{R}$. Hence, the series for f' has the same radius of convergence as f. Let $z \in \mathbb{C}$, such that $|z| < R$. Choose δ such that $|z| + \delta < R$. Then, for $|h| < \delta$, we want to show that $\frac{f(z+h)-f(z)}{h} - \sum_{n=1}^{\infty} na_n z^{n-1} \to 0$ as $h \to 0$. We have

$$\begin{aligned} &\left|\sum_{n=0}^{\infty} \frac{a_n(z + h)^n - a_n z^n}{h} - na_n z^{n-1}\right| \\ &\leq \sum_{n=0}^{\infty} |a_n| \left|\frac{1}{h}\left(\sum_{k=0}^{n} \binom{n}{k} z^{n-k}h^k - z^n\right) - nz^{n-1}\right| \\ &= \sum_{n=0}^{\infty} |a_n| \left|\sum_{k=2}^{n} \binom{n}{k} h^{k-1} z^{n-k}\right|. \end{aligned}$$

Since the first two terms cancel. Now, pulling out the first two terms of the binomial coefficient, we have

$$\begin{aligned}
&\sum_{n=0}^{\infty} |a_n| \left| \sum_{k=2}^{n} \binom{n}{k} h^{k-1} z^{n-k} \right| \\
&\leq \sum_{n=0}^{\infty} |a_n| (n(n-1)) \sum_{k=2}^{n} \binom{n-2}{k-2} |h|^{k-2} |z|^{n-k} |h| \\
&\leq |h| \sum_{n=0}^{\infty} |a_n| n(n-1)(|z|+|h|)^{n-2} \\
&\leq |h| \sum_{n=0}^{\infty} |a_n| n(n-1)(|z|+\delta)^{n-2}
\end{aligned}$$

Since $|z|+\delta < R$, the series $\sum_{n=0}^{\infty} |a_n| n(n-1)(|z|+\delta)^{n-2}$ converges. Hence, we have that this expression goes to 0 as $h \to 0$. Hence, f is holomorphic with the derivative given by the termwise derivative of the power series. □

Example 1.19:

The exponential function is an entire function and its derivative is given by

$$\frac{d}{dz} e^z = \sum_{n=1}^{\infty} \frac{n z^{n-1}}{n!} = \sum_{n=1}^{\infty} \frac{z^{n-1}}{(n-1)!} = \sum_{m=0}^{\infty} \frac{z^m}{m!} = e^z.$$

Example 1.20: Trigonometric Functions

We define the complex sine and cosine functions as

$$\cos(z) := \frac{e^{iz} + e^{-iz}}{2} = \sum_{n=0}^{\infty} \frac{(-1)^n z^{2n}}{(2n)!},$$

$$\sin(z) := \frac{e^{iz} - e^{-iz}}{2i} = \sum_{n=0}^{\infty} \frac{(-1)^n z^{2n+1}}{(2n+1)!}.$$

Both of these functions are entire.

As a warning, the complex sine and cosine functions are not bounded, unlike their real counterparts!

Example 1.21: Geometric Series

The geometric series is given by

$$\sum_{n=0}^{\infty} z^n = \frac{1}{1-z},$$

which converges for $|z| < 1$.

1.7 Complex Line Integrals

We like to recap some definitions.

Definition 1.22: Parametrized Curve

A PARAMETRIZED CURVE in $\mathbb{C}$ is a continuous function $\gamma : [a, b] \to \mathbb{C}$.

We call a curve SMOOTH if its derivative $\gamma'(t)$ exists and $\forall t \in [a, b]$, $\gamma'(t)$ is continuous and $\gamma'(t) \neq 0$.

A PIECEWISE SMOOTH curve is a curve γ such that $\exists$ points $a = a_0 < a_1 < \ldots < a_n = b$ such that $\gamma|_{[a_i, a_{i+1}]}$ is smooth for all $i = 0, \ldots, n-1$.

A CLOSED CURVE is a curve γ such that $\gamma(a) = \gamma(b)$. A curve is SIMPLE if it is not self-intersecting.

In the following we will always assume that a curve is piecewise smooth.

Definition 1.23: Reparametrization

A REPARAMETRIZATION of a curve $\gamma : [a, b] \to \mathbb{C}$ is a curve

$$\tilde{\gamma} : [c, d] \to \mathbb{C},$$

such that there exists a continuously differentiable function $\sigma : [c, d] \to [a, b]$ which is bijective and $\sigma'(t) > 0$ for all $t \in [c, d]$ and $\tilde{\gamma}(t) = \gamma(\sigma(t))$.

Example 1.24: Parametrized Line Segment

Given 2 points $z_1, z_2 \in \mathbb{C}$, the path

$$\gamma : [0, 1] \to \mathbb{C}, t \mapsto \gamma(t) = (1-t) z_1 + t z_2,$$

is the line segment between z_1 and z_2.

Example 1.25: Parametrized Circle

A circle of center $z_0 \in \mathbb{C}$ and radius $r > 0$ can be parametrized by

$$\gamma : [0, 2\pi] \to \mathbb{C}, t \mapsto (t) = z_0 + r e^{it}.$$

We want to define path integrals of complex functions. Recall that if g is a real-valued continuous function of a real variable, then it is Riemann-integrable. If g is complex valued $g : [a, b] \to \mathbb{C}, g(t) = u(t) + iv(t)$, then we define

$$\int_a^b g(t) \mathrm{d}t = \int_a^b u(t) \mathrm{d}t + i \int_a^b v(t) \mathrm{d}t.$$

Definition 1.26: Line Integral

Suppose $\gamma : [a, b] \to \Omega \subset \mathbb{C}$ is a piecewise smooth curve and $f : \Omega \to \mathbb{C}$ is defined and continuous on the curve. Then the LINE INTEGRAL of f along γ is defined by

$$\int_\gamma f(z) \mathrm{d}z = \int_a^b f(\gamma(t)) \gamma'(t) \mathrm{d}t.$$

Lec 5

The line integral is independet of the parametrization of the curve. The proof of this is analogous to analysis II.

Proposition 1.27:

Given $f, g : \Omega \to \mathbb{C}$ continuous, γ_1, γ_2 2 piecewise smooth curves in Ω. Then

1. $\int_\gamma (af + bg)dz = a\int_\gamma f dz + b\int_\gamma g dz$ for $a, b \in \mathbb{C}$
2. If γ^- is the curve γ with reverse orientation, then $\int_{\gamma^-} f dz = -\int_\gamma f dz$
3. $\int_{\gamma_1+\gamma_2} f\mathrm{d}z = \int_{\gamma_1} f\mathrm{d}z + \int_{\gamma_2} f\mathrm{d}z$.
4. $\left|\int_\gamma f dz\right| \leq (\sup_{z\in\gamma} |f(z)|) \cdot L(\gamma)$, where $L(\gamma)$ is the length of the curve γ.

Here, the length of a piecewise smooth curve $\gamma : [a, b] \to \mathbb{C}$ is defined as

$$L(\gamma) = \sum_{i=0}^{n-1} \int_{a_i}^{a_{i+1}} |\gamma'(t)|\mathrm{d}t.$$

Definition 1.28: Primitive

Let $f : \Omega \to \mathbb{C}$. A PRIMITIVE for f is a function F that is holomorphic on Ω and satisfies $F'(z) = f(z)$ for all $z \in \Omega$.

Existance of a primitive gives the following result.

Theorem 1.29: Fundamental Theorem for Line Integrals

Let $f : \Omega \to \mathbb{C}$ be continuous on an open set Ω. If f has a primitive F in Ω, and $\gamma : [a, b] \to \Omega$ is a curve, which starts at z_1 and ends at z_2, then

$$\int_\gamma f\mathrm{d}z = F(z_2) - F(z_1).$$

Proof. Let $F = U(x, y) + iV(x, y)$, where U, V are real valued functions and $\gamma : [a, b] \to \mathbb{C}$. We define a function

$$\begin{aligned} G : [a, b] &\longrightarrow \mathbb{C} \\ t &\longmapsto G(t) = F(\gamma(t)) = F((x(t), y(t))). \end{aligned}$$

We have to check the compatibility of the derivative of G and the complex derivative of F. We have

$$G'(t) = \frac{\partial U}{\partial x}\frac{dx}{dt} + \frac{\partial U}{\partial y}\frac{dy}{dt} + i\left(\frac{\partial V}{\partial x}\frac{dx}{dt} + \frac{\partial V}{\partial y}\frac{dy}{dt}\right).$$

Since F is holomorphic, we have the Cauchy-Riemann equations and can write

$$G'(t) = U_x(x, y)x'(t) - V_x(x, y)y'(t) + i\left(V_x(x, y)x'(t) + U_x(x, y)y'(t)\right).$$

Regrouping the terms, we get

$$G'(t) = \underbrace{(U_x + iV_x)}_{F'(x,y)}(x'(t) + iy'(t)) = F'(\gamma(t))\gamma'(t) = f(\gamma(t))\gamma'(t).$$

This shows, that

$$\int_\gamma f\mathrm{d}z = \int_a^b f(\gamma(t))\gamma'(t)dt = \int_a^b G'(t)dt.$$

This is now a real line integral and we can apply the fundamental theorem of calculus to get

$$\int_\gamma f\mathrm{d}z = G(b) - G(a) = F(\gamma(b)) - F(\gamma(a)) = F(z_2) - F(z_1).$$

□

Corollary 1.30:

If γ is a closed curve in Ω, f is continuous and has a primitive in Ω, then

$$\int_\gamma f\mathrm{d}z = 0.$$

Corollary 1.31:

If $f : \Omega \to \mathbb{C}$ is holomorphic, where Ω is open and connected. If $f' = 0$, then f is constant.

Proof. We want to show, that for any two points $z, w \in \Omega$ $f(z) = f(w)$. Since Ω is open and connected, there is a polygonal path $\gamma : [a, b] \to \Omega$ connecting z and w. Since f is holomorphic, it is the primitive for its derivative f'. By the fundamental theorem of line integrals, we have

$$\int_\gamma f'\mathrm{d}z = f(w) - f(z) = 0.$$

□

Example 1.32:

Given $f : \mathbb{C} \setminus \{0\} \to \mathbb{C}, f(z) = \frac{1}{z}$. Then f has no primitive on $\mathbb{C} \setminus \{0\}$.

Let γ be the unit circle, $\gamma : [0, 2\pi] - > \mathbb{C}, t \mapsto e^{it}$. Then

$$\int_\gamma f\mathrm{d}z = \int_0^{2\pi} \frac{1}{e^{it}} ie^{it}dt = 2\pi i \neq 0.$$

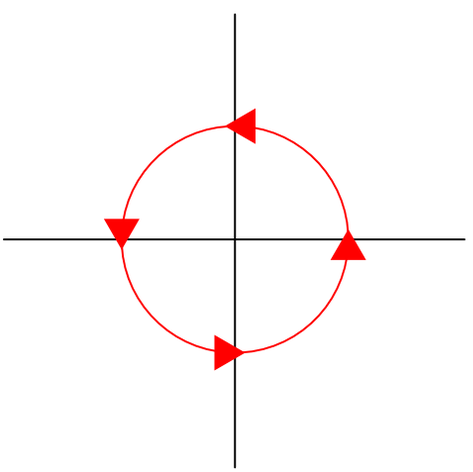


Figure 2: parametrization of the unit circle

2 Cauchy's Theorem on a disc

Lec 6

If f is holomorphic on a disk D, γ is a closed curve such that γ and its interior are contained in D, then $\int_\gamma f\mathrm{d}z = 0$. This has many application.

Theorem 2.1: Goursat's Theorem

Let $\Omega \subset \mathbb{C}$ be open, $T \subset \Omega$ is a triangle whose interior is contained in Ω. Let $f : \Omega \to \mathbb{C}$ be holomorphic. Then

$$\int_T f\mathrm{d}z = 0.$$

For the proof, we will make use the following proposition.

Proposition 2.2: Compact nested sets

If $K_1 \supset K_2 \supset K_3 \supset \ldots$ is a sequence of compact nested sets in $\mathbb{C}$ with the property, that the diameter of the sets tends to zero, then $\exists$ a unique point $z_0 \in \mathbb{C}$ such that $z_0 \in K_n$ for all $n \in \mathbb{N}$.

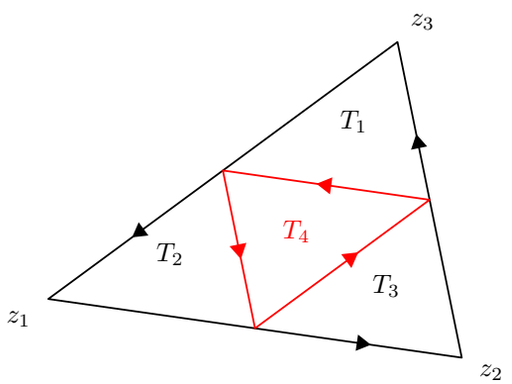


Figure 3: Proof of Goursats Theorem

Proof. Note that a triangle is a closed curve, which is the union of three line segments. If T has three corners z_1, z_2, z_3, then we can write $T =< z_1, z_2, z_3 >= \alpha_1 + \alpha_2 + \alpha_3$, where α_i are the line segments connecting the corners.

$$\begin{aligned}\alpha_1 &= z_1 + (t-0)(z_2 - z_1), \quad t \in [0,1]\\ \alpha_2 &= z_2 + (t-1)(z_3 - z_2), \quad t \in [1,2]\\ \alpha_3 &= z_3 + (t-2)(z_1 - z_3), \quad t \in [2,3].\end{aligned}$$

We also write $\Delta = \{z \in \mathbb{C} | z = t_1 z_1 + t_2 z_2 + t_3 z_3\}$, such that $T = \partial\Delta$. We will inductively construct a sequence of triangular paths. We define

$$T^{(n)} =< z_1^{(n)}, z_2^{(n)}, z_3^{(n)} >$$

as follows:

$$T^{(0)} =< z_1, z_2, z_3 > .$$

When $T^{(n)}$ is defined, we define $T^{(n+1)}$ as one of the 4 triangular paths

$$\begin{aligned}T_1^{(n)} &= \left\langle \frac{z_1^{(n)} + z_2^{(n)}}{2}, z_2^{(n)}, \frac{z_2^{(n)} + z_3^{(n)}}{2} \right\rangle\\ T_2^{(n)} &= \left\langle z_1^{(n)}, \frac{z_1^{(n)} + z_2^{(n)}}{2}, \frac{z_1^{(n)} + z_3^{(n)}}{2} \right\rangle\\ T_3^{(n)} &= \left\langle \frac{z_1^{(n)} + z_3^{(n)}}{2}, \frac{z_2^{(n)} + z_3^{(n)}}{2}, z_3^{(n)} \right\rangle\\ T_4^{(n)} &= \left\langle \frac{z_1^{(n)} + z_2^{(n)}}{2}, \frac{z_1^{(n)} + z_3^{(n)}}{2}, \frac{z_2^{(n)} + z_3^{(n)}}{2} \right\rangle.\end{aligned}$$

At each step, we partition Δ into 4 new ones by taking midpoints of the sides of T.

We have, that

$$\begin{aligned}\int_{T^{(n)}} f\mathrm{d}z &= \int_{T_1^{(n)}} f\mathrm{d}z + \int_{T_2^{(n)}} f\mathrm{d}z + \int_{T_3^{(n)}} f\mathrm{d}z + \int_{T_4^{(n)}} f\mathrm{d}z\\ &\le 4 \max_{i=1,2,3,4} |\int_{T_i^{(n)}} f\mathrm{d}z|.\end{aligned}$$

We choose $T^{(n+1)}$ as one of $T_i^{(n)}$ such that

$$\left|int_{T^{(n)}} f\mathrm{d}z\right| \le 4\left|\int_{T^{(n+1)}} f\mathrm{d}z\right|.$$

But now recursively, we get

$$\left|\int_{T^{(n)}} f\mathrm{d}z\right| \le 4^n \left|\int_{T^{(0)}} f\mathrm{d}z\right|.$$

The closed filled triangles $\Delta^{(n)}$ corresponding to $T^{(n)}$ are nested compact sets, i.e.

$$\Delta = \Delta^{(0)} \supset \Delta^{(1)} \supset \Delta^{(2)} \supset \ldots.$$

Moreover, if $\mathrm{diam}(\Delta) := \sup\{|z-w| : z, w \in \Delta\} =: d$, and d_n and P_n are the diameters and perimeters of the triangles $\Delta^{(n)}$, then $d_n = \frac{d}{2^n}$ and $P_n = \frac{P}{2^n}$.

Hence, $\mathrm{diam}(\Delta^{(n)}) \to 0$ as $n \to \infty$. By compact nested sets, there is a unique point $z_0 \in \mathbb{C}$ such that $z_0 \in \Delta^{(n)}$ for all $n \in \mathbb{N}$.

Going back to our integrals, we have, since f is holomorphic at z_0,i.e.

$$f(z) = f(z_0) + f'(z_0)(z - z_0) + \epsilon(z), \quad \text{where } \epsilon(z) \to 0 \text{ as } z \to z_0.$$

Notice, that $\epsilon(z)$ is continous at z_0 and $\lim_{z\to z_0} \epsilon(z) = 0$.

$$\begin{aligned}I &:= \left|\int_{T^{(0)}} f\mathrm{d}z\right| \le 4^n \left|\int_{T^{(n)}} f\mathrm{d}z\right|\\ &= 4^n \left|\int_{T^{(n)}} f(z_0) + f'(z_0)(z - z_0)\mathrm{d}z + \int_{T^{(n)}} \epsilon(z)\mathrm{d}z\right|.\end{aligned}$$

The first integral is zero, since

$$G(z) = f(z_0)z + \frac{f'(z_0)}{2}z^2$$

is a primitive of $f(z_0) + f'(z_0)(z - z_0)$. Hence, we have

$$I \le 4^n \left|\int_{T^{(n)}} \epsilon(z)\mathrm{d}z\right|.$$

For a given $\varepsilon > 0$, choose an open disc $D_\delta(z_0) \subset \Omega$ such that

$$|\epsilon(z)| < \varepsilon|z - z_0|, \quad \forall z \in D_\delta(z_0).$$

Because $\mathrm{diam}(\Delta^{(n)}) \to 0$, there exists $N \in \mathbb{N}$ such that $d_n < \delta$ for all $n \ge N$. Hence, we have $z_0 \in \Delta^{(n)}$ and $\forall z \in \Delta^{(n)}$, $|z - z_0| < d_n < \delta$. Therefore, we have

$$\Delta^{(n)} \subset D_\delta(z_0), \quad \forall n \ge N.$$

In total,

$$\begin{aligned}|I| &\le 4^n \left|\int_{T^{(n)}} \epsilon(z)\mathrm{d}z\right|\\ &\le 4^n \int_{T^{(n)}} |\epsilon(z)||dz|\\ &\le 4^n \varepsilon \int_{T^{(n)}} |z - z_0||dz|\\ &\le 4^n \varepsilon d_n L(T^{(n)})\\ &= 4^n \varepsilon \frac{d}{2^n}\frac{P}{2^n} = \varepsilon dP.\end{aligned}$$

Letting $\varepsilon \to 0$, we get $|I| = 0$, which proves the theorem. □

Corollary 2.3:

If $f : \Omega \to \mathbb{C}$ is holomorphic, the open set Ω contains a Recatangle $\mathcal{R}$ and its interior R such that $\partial R = \mathcal{R}$, then

$$\int_{\mathcal{R}} f\mathrm{d}z = 0.$$

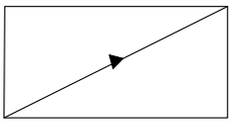


Figure 4: rectangle

Proof. We split the rectangle into two triangles by drawing a diagonal. Then we can apply Goursat's theorem to each triangle and get the result. □

Theorem 2.4: Goursat

If a function f is continuous in an open subset $\Omega \subset \mathbb{C}$ and holomorphic in $\Omega \setminus \{z_0\}$, then

$$\int_{\mathcal{R}} f(z)\mathrm{d}z = 0.$$

For every closed rectangle $\mathcal{R} \subset \Omega$.

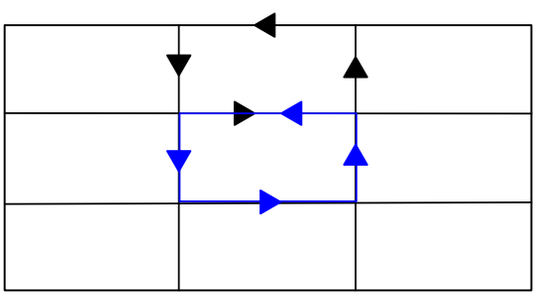


Figure 5: Goursat's theorem for a rectangle

Proof. For a closed rectangle $R \subset \Omega$, we assume $z_0 \in \mathcal{R}$, otherwise the conclusion follows from the previous version.

Given a positive integer n, we divide $\mathcal{R}$ into n^2 congruent ones. Once again, it is clear, that

$$\int_{\mathcal{R}} f(z)\mathrm{d}z = \sum_{k=1}^{n}\sum_{l=1}^{n}\int_{\mathcal{R}_{k,l}} f(z)\mathrm{d}z.$$

If $z_0 \notin \mathcal{R}_{k,l}$, then $\int_{\mathcal{R}_{k,l}} f(z)\mathrm{d}z = 0$ by the previous version of Goursat's theorem. If $z_0 \in \mathcal{R}_{k,l}$, then we have

$$\int_{\mathcal{R}_{k,l}} f(z)\mathrm{d}z \leq M\text{perimeter}(\mathcal{R}_{k,l}) = \frac{ML}{n}.$$

Where, $M = \max_{z\in\mathcal{R}} |f(z)|$ and L is the perimeter of $\mathcal{R}$.

But notice, that we have at most four rectangles containing z_0, hence letting $n \to \infty$, we get

$$\left|\int_{\mathcal{R}} f(z)\mathrm{d}z\right| \leq 4\frac{ML}{n} \to 0.$$

□

To prove Cauchy's theorem in a disc, we need the following theorem.

Theorem 2.5:

A holomorphic function on an open disc, has a primitive in that disc.

Instead, we'll prove the following version

Theorem 2.6:

Let D be an open disc in $\mathbb{C}$, f continuous on D with the property that $\int_{\mathcal{R}} f(z)\mathrm{d}z = 0$ for every closed rectangle $R \subset D$, whose sides are parallel to the coordinate axes. Then f has a primitive in D.

Proof. f is continuous in the disc. Let $z_0 = x_0 + iy_0 \in D$. Let this be the center of the disk. For an arbitrary point $z \in D$, such that $z \neq z_0$, let

$$z_1 = x + iy_0, \quad z_2 = x_0 + iy.$$

By assumption,

$$\int_{z_0}^{z_1} f(w)\mathrm{d}w + \int_{z_1}^{z} f(w)\mathrm{d}w + \int_{z}^{z_2} f(w)\mathrm{d}w + \int_{z_2}^{z_0} f(w)\mathrm{d}w = 0.$$

We define $F : D \to \mathbb{C}$ as: For $z \in D$, let

$$F(z) := \int_{z_0}^{z_2} f(w)\mathrm{d}w + \int_{z_2}^{z} f(w)\mathrm{d}w.$$

As shown, we can also write

$$F(z) = \int_{z_0}^{z_1} f(w)\mathrm{d}w + \int_{z_1}^{z} f(w)\mathrm{d}w.$$

We parametrize the line segments

$$\begin{aligned} F(z) &:= i\int_{y_0}^{y} f(x_0 + it)\mathrm{d}t + \int_{x_0}^{x} f(t + iy)\mathrm{d}t \\ &= \int_{x_0}^{x} f(t + iy_0)\mathrm{d}t + i\int_{y_0}^{y} f(x + it)\mathrm{d}t. \end{aligned}$$

Note that for the first equation, the first integral is independent of x, and the second one, when we differentiate with respect to x, we get by the fundamental theorem of calculus, that

$$\frac{\partial F}{\partial x} = f(x + iy).$$

Doing the same for the second equation, we get

$$\frac{\partial F}{\partial y} = if(x + iy).$$

Combining the equations, we have

$$F_x = f(z), F_y = if(z) \Rightarrow f(z) = F_x = -iF_y.$$

Exercise: Check the CR equations for F and conclude that F is holomorphic in D and $F'(z) = f(z)$. □

Note that this theorem as a corollary gives us Cauchy's theorem in a disc

Theorem 2.7: Cauchy's theorem in a disc

Let $D \subset \mathbb{C}$ be an open disc, $f : D \to \mathbb{C}$ holomorphic (or more generally continuous on D and holomorphic on $D \setminus \{z_0\}$ for some $z_0 \in D$). Then

$$\int_{\gamma} f(z)\mathrm{d}z = 0,$$

for every closed, piecewise smooth curve γ in D, whose interior is contained in D.

Proof. Suppose f is holomorphic on $D \setminus \{z_0\}$ for some $z_0 \in D$. Then by Goursat's theorem, the integral of f along any rectangle is zero. In particular also the ones whose sides are parallel to the coordinate

axes. Hence, by the previous theorem, f has a primitive in D. By the fundamental theorem of line integrals, we have

$$\int_\gamma f(z)\mathrm{d}z = 0.$$

Lec 7

□

Corollary 2.8:

Suppose f is holomorphic in an open set $\Omega \subset \mathbb{C}$ containing a circle C and its interior. Then

$$\int_C f(z)\,dz = 0.$$

Proof. Since Ω is open, there exists a disk D centered at the center of C such that $D \subset \Omega$. By Cauchy's theorem, we have

$$\int_C f(z)\,dz = 0.$$

□

Recall that in showing that f has a primitive in a disc, we used a polygonal path to define

$$F(z) = \int_{\gamma_z} f(w)\,dw.$$

In the corollary, circles, rectangles and other polygons are curves whose interior are easy to define. We call them TOY CONTOURS. Inside each of these toy contours, we can define a primitive of f by integrating along a path from a fixed point to z.

Example 2.9:

We claim that $\int_R \frac{1}{z-z_0}\,dz = 2\pi i$ for any Rectangle with center at z_0.

Recall, that $\int_C \frac{1}{z-z_0}\,dz = 2\pi i$ for any circle C with center at z_0. Now, consider a large circle D with z_0 on its boundary. Note that $f = \frac{1}{z-z_0}$ is holomorphic in the interior of D. Now the curve $\beta_1+\gamma_1$ is a closed curve in the disc. So by Cauchy's theorem, we have

$$\int_{\beta_1} \frac{1}{z-z_0}\,dz + \int_{\gamma_1} \frac{1}{z-z_0}\,dz = 0.$$

Doing the same for all other sides of the rectangle, we have

$$2\pi i = \int_C \frac{1}{z-z_0}\,dz = \int_R \frac{1}{z-z_0}\,dz = \int_R \frac{1}{z-z_0}\,dz.$$

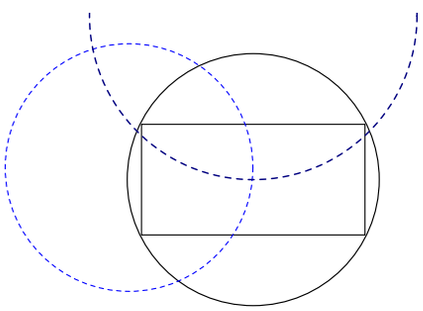


Figure 6: Transforming the curve

Example 2.10:

Show that $\int_0^\infty \cos(x^2)\,dx = \int_0^\infty \sin(x^2)\,dx = \frac{\sqrt{2\pi}}{4}$.

Solution. We want to show that $\lim_{r\to\infty}\int_0^r \cos(x^2)\,dx = \lim_{r\to\infty}\int_0^r \sin(x^2)\,dx = \frac{\sqrt{2\pi}}{4}$. Note, that $\cos(x^2)+i\sin(x^2) = e^{ix^2}$. ie we want to show that

$$\lim_{r\to\infty}\int_0^r e^{ix^2}\,dx = (1+i)\frac{\sqrt{2\pi}}{4}.$$

This leads to the holomorphic function $f(z) = e^{iz^2}$. Considering the line γ_1 from 0 to r, we have

$$\int_{\gamma_1} f(z)\,dz = \int_0^r e^{ix^2}\,dx.$$

We now consider the line γ_2 from 0 to $re^{i\pi/4}$ along constant radius. Lastly γ_3 is the line from $re^{i\pi/4}$ to 0. Since f is holomorphic everywhere, and $\gamma_1+\gamma_2+\gamma_3$ is a closed curve, we have

$$\int_{\gamma_1+\gamma_2+\gamma_3} f(z)\,dz = 0.$$

We will use the following parametrizations

$$\begin{aligned}
\gamma_1(t) &= t, && 0 \le t \le r\\
\gamma_2(t) &= re^{it}, && 0 \le t \le \frac{\pi}{4}\\
\gamma_3(t) &= (r-t)e^{i\pi/4}, && 0 \le t \le r.
\end{aligned}$$

Hence we have

$$\begin{aligned}
\int_0^r e^{ix^2}\,dx &= -\int_{\gamma_2} f(z)\,dz - \int_{\gamma_3} f(z)\,dz\\
&= -\int_0^{\pi/4} e^{i(r^2e^{2it})} ire^{it}\,dt - \int_0^r e^{i((r-t)^2e^{i\pi/2})}e^{i\pi/4}\,dt
\end{aligned}$$

$$\int_0^\infty e^{ix^2}\,dx = -\lim_{r\to\infty}\int_0^{\pi/4} e^{i(r^2e^{2it})} ire^{it}\,dt - \frac{\sqrt{\pi}}{2}e^{i\pi/4}.$$

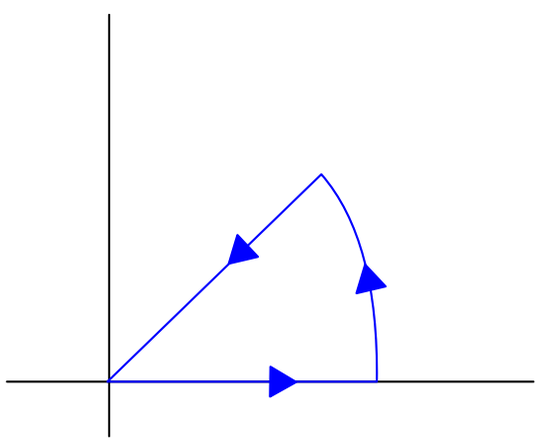


Figure 7: Curve used for the Fresnel integrals

Lec 8

Theorem 2.11: Cauchy Integral Formula

Suppose $f : \Omega \to \mathbb{C}$ is holomorphic on an open set $\Omega \subset \mathbb{C}$. Suppose Ω contains the closure of a disc D. If C denotes the boundary of D traced positively (in counterclockwise direction), then

$$f(z) = \frac{1}{2\pi i}\int_C \frac{f(w)}{w-z}\,dw \quad \text{for all } z \in D.$$

Proof. Let $z \in D_r(z_0)$, $\overline{D_r(z_0)} \subset \Omega$. Let $\varepsilon > 0$ such that

$$D_{r+\varepsilon}(z_0) \subset \Omega.$$

For $w \in D_{r+\varepsilon}(z_0)$, define

$$g(w) := \begin{cases} \frac{f(w)-f(z)}{w-z} & w \neq z \\ f'(z) & w = z \end{cases}.$$

Then $g : D_{r+\varepsilon}(z_0) \to \mathbb{C}$ is continuous on the disc and holomorphic on $D_{r+\varepsilon}(z_0) \setminus \{z\}$. So by Cauchy's theorem,

$$\int_{C_r(z_0)} g(w)\,dw = 0 \Rightarrow \int_{C_r(z_0)} \frac{f(w)-f(z)}{w-z}\,dw = 0.$$

Reformulating, we get

$$\int_{C_r(z_0)} \frac{f(w)}{w-z}\,dw = f(z)\int_{C_r(z_0)} \frac{1}{w-z}\,dw.$$

We claim, that $I = \int_{C_r(z_0)} \frac{1}{w-z}\,dw = 2\pi i$.

The standard parametrization of $C_r(z_0)$ is given by $\gamma(t) = z_0 + re^{it}$, $t \in [0, 2\pi]$. Then

$$\begin{aligned} I &= \int_0^{2\pi} \frac{ire^{it}}{re^{it}+z_0-z}\,dt \\ &= \int_0^{2\pi} \frac{ie^{it}}{e^{it} - \underbrace{\frac{z-z_0}{r}}_{:=a}}\,dt. \end{aligned}$$

Note, that $|a| < 1$ since $z \in D_r(z_0)$. We can rewrite the integrand as

$$\frac{ie^{it}}{e^{it}-a} = \frac{i}{1-ae^{-it}} = i\sum_{n=0}^{\infty} a^n e^{-int}.$$

By uniform convergence of the series, we can interchange the sum and the integral and since the only non-zero integral is the one for $n = 0$, we get

$$I = i\int_0^{2\pi} 1\,dt = 2\pi i.$$

Alternatively, we can use an offset parametrization of the circle

$$\tilde{\gamma}(s) = z + \rho(s)e^{is}, \quad s \in [0, 2\pi], \quad \rho(s) = |\gamma(t(s)) - z|.$$

Now, $\tilde{\gamma}'(s) = \rho'(s)e^{is} + i\rho(s)e^{is}$. The rest is left as an exercise. □

Theorem 2.12: Holomorphic functions are analytic

Suppose f is holomorphic in an open set Ω. Let $z_0 \in \Omega$ and $r > 0$ such that $\overline{D_r(z_0)} \subset \Omega$. Then f has a power series representation at z_0

$$f(z) = \sum_{n=0}^{\infty} a_n(z-z_0)^n \quad \forall z \in D_r(z_0).$$

The coefficients are given by

$$a_n = \frac{f^{(n)}(z_0)}{n!} = \frac{1}{2\pi i}\int_{C_r(z_0)} \frac{f(w)}{(w-z_0)^{n+1}}\,dw.$$

Proof. Let z_0, r be as in the theorem. Fix some $s \in (0, r)$ and let $C_s(z_0)$ be the boundary of $D_s(z_0)$ traced positively. Then $\gamma = C_s(z_0)$ is contained in Ω. By Cauchy's integral formula, we have

$$f(z) = \frac{1}{2\pi i}\int_{C_s(z_0)} \frac{f(w)}{w-z}\,dw \quad \forall z \in D_s(z_0).$$

The trick is now, to write

$$\frac{1}{w-z} = \frac{1}{w-z_0-(z-z_0)} = \frac{1}{w-z_0}\cdot\left(\frac{1}{1-\frac{z-z_0}{w-z_0}}\right).$$

This gives us a geometric series expansion, since $|z-z_0| < s < |w-z_0|$ for all $w \in C_s(z_0)$. Hence,

$$\frac{1}{w-z} = \frac{1}{w-z_0}\sum_{n=0}^{\infty}\left(\frac{z-z_0}{w-z_0}\right)^n.$$

We can interchange the sum and the integral, since the series converges uniformly on $C_s(z_0)$. This gives us

$$\begin{aligned} f(z) &= \frac{1}{2\pi i}\int_\gamma f(w)\sum_{n=0}^{\infty}(z-z_0)^n(w-z_0)^{-n-1}\,dw \\ &= \frac{1}{2\pi i}\sum_{n=0}^{\infty}(z-z_0)^n\int_\gamma \frac{f(w)}{(w-z_0)^{n+1}}\,dw \\ &= \sum_{n=0}^{\infty} a_n(z-z_0)^n \quad \text{with } a_n = \frac{1}{2\pi i}\int_\gamma \frac{f(w)}{(w-z_0)^{n+1}}\,dw. \end{aligned}$$

So we have that for any z in $D_s(z_0)$ f has a power series representation. Since we've seen that power series in their disc of convergence are holomorphic and can be differentiated termwise, we get

$$f'(z) = \sum na_n(z-z_0)^{n-1} \quad \forall z \in D_s(z_0).$$

We can repeat to find

$$f^{(n)}(z) = \sum_{k=n}^{\infty} k(k-1)\cdots(k-n+1)a_k(z-z_0)^{k-n} \quad \forall z \in D_s(z_0).$$

Hence, f can be differentiated infinitely often.

For $z \in D_s(z_0)$, evaluating the power series at z_0 gives

$$a_0 = f(z_0), \quad a_1 = f'(z_0), \quad a_n = \frac{f^{(n)}(z_0)}{n!}.$$

□

Corollary 2.13: Cauchy Integral Formula for Derivatives

If $f \in \mathscr{H}(\Omega)$, then f is infinitely differentiable in Ω. Moreover if $z_0 \in \Omega$ such that $\exists r > 0$ with $\overline{D_r(z_0)} \subset \Omega$, then for any $z \in D_r(z_0)$

$$f^{(n)}(z) = \frac{n!}{2\pi i}\int_{C_r(z)} \frac{f(w)}{(w-z)^{n+1}}\,dw.$$

Proof. The fact that f is infinitely differentiable follows from the power series representation. To prove the integral formula, To prove the integral formula, we can use induction on n. $n = 0$ is simply the Cauchy integral formula. Suppose

$$f^{(n-1)}(z) = \frac{(n-1)!}{2\pi i}\int_{C_r(z)} \frac{f(w)}{(w-z)^n}\,dw.$$

Choose h small, so that $z, z+h \in D_r(z_0)$. Then

$$\begin{aligned} &\frac{f^{(n-1)}(z+h) - f^{n-1}(z)}{h} \\ &= \frac{(n-1)!}{2\pi i}\int_{C_r(z)} \frac{f(w)}{h}\left(\frac{1}{(w-z-h)^n} - \frac{1}{(w-z)^n}\right)dw. \end{aligned}$$

Using $(a^n - b^n) = (a-b)(a^{n-1} + a^{n-2}b + \cdots + b^{n-1})$, we can rewrite the integrand with $a = \frac{1}{(w-z-h)}$ and $b = \frac{1}{(w-z)}$ to get

$$\frac{a-b}{h} = \frac{1}{h}\left(\frac{h}{(w-z-h)(w-z)}\right) \xrightarrow{h\to 0} \frac{1}{(w-z)^2},$$

and

$$a^{n-1} + a^{n-2}b + \cdots + b^{n-1} \xrightarrow{h\to 0} n\frac{1}{(w-z)^{n-1}}.$$

Hence,

$$\lim_{h\to 0}\frac{f^{(n-1)}(z+h) - f^{n-1}(z)}{h} = \frac{n!}{2\pi i}\int_{C_r(z)} \frac{f(w)}{(w-z)^{n+1}}\,dw.$$

□

Corollary 2.14: Cauchy Inequalities

With the assumptions of the previous corollary, we have

$$\left|f^{(n)}(z_0)\right| \le \frac{n!}{r^n}\sup_{|z-z_0|=r}|f(z)|.$$

Proof. We have seen, $|a_n| = \left|\frac{f^{(n)}(z_0)}{n!}\right| = \left|\frac{1}{2\pi i}\int_{C_r(z_0)} \frac{f(w)}{(w-z_0)^{n+1}}\,dw\right|$. Hence,

$$|n!a_n| \le \frac{n!}{2\pi}\int_{C_r(z_0)} \frac{|f(w)|}{|w-z_0|^{n+1}}\,|dw|$$
$$\le n!\frac{\sup_{|z-z_0|=r}|f(z)|}{r^n}.$$

□

Theorem 2.15: Liouville's Theorem

Suppose $f:\mathbb{C}\to\mathbb{C}$ is entire and bounded. Then f is constant.

Proof. Since $\mathbb{C}$ is open and connected, to show that f is constant, it suffices to show that $f'=0$. Let $z_0\in\mathbb{C}$ and $r>0$. Then $D_r(z_0)\subset\mathbb{C}$ for any $r>0$. Using the Cauchy Inequality for the first derivative, we get

$$|f'(z_0)| \le \frac{1}{r}\sup_{|z-z_0|=r}|f(z)| \le \frac{M}{r}.$$

Since f is bounded on $\mathbb{C}$. Now let $r\to\infty$, to get $|f'(z_0)|=0$. Since z_0 was arbitrary, we have $f'=0$ on $\mathbb{C}$, hence f is constant. □

Corollary 2.16: Fundamental Theorem of Algebra

If $p(z)=a_nz^n+\cdots+a_0$ is a polynomial of degree $n\ge 1$, with $a_n\neq 0$, then f has n roots in $\mathbb{C}$ counting multiplicity.

If $\omega_1,\dots,\omega_n$ are the roots of p, then we can factor p as

$$p(z)=a_n(z-\omega_1)\cdots(z-\omega_n).$$

Proof. We'll first show that p has at least one root. Suppose not, then $Q(z)=\frac{1}{p(z)}$ is holomorphic everywhere. If we can show that Q is bounded, then by Liouville's theorem, Q is constant, which implies that p is constant, which is a contradiction.

Let $z\neq 0$. Then

$$\begin{aligned}|P(z)| &= |a_nz^n+\cdots+a_0|\\ &\ge |a_n||z|^n-|a_{n-1}||z|^{n-1}-\cdots-|a_0|\\ &= |z|^n\left(|a_n|-\frac{|a_{n-1}|}{|z|}-\cdots-\frac{|a_0|}{|z|^n}\right).\end{aligned}$$

This shows that for $|Q(z)|$ goes to zero as $|z|\to\infty$. Hence, $\exists R>0$ such that $|Q(z)|\le 1$ when $|z|>R$. On the other hand, $Q(z)$ is continuous, hence also bounded on the compact set $\overline{D_R(0)}$. Hence, Q is bounded on $\mathbb{C}$.

Hence, p has a zero in $\mathbb{C}$, say at ω_1. Then we can write

$$p(z)=a_n(z-\omega_1+\omega_1)^n+\cdots+a_0=b_n(z-\omega_1)^n+\cdots+b_0.$$

Since $p(\omega_1)=0$, we have $b_0=0$. Hence, we can factor out $(z-\omega_1)$ to get

$$p(z)=(z-\omega_1)q(z).$$

Applying the same argument to q, we can find a root ω_2 of q, and factor out $(z-\omega_2)$. Repeating this process, we can find n roots of p, counting multiplicity. □

Lec 9

Remark 2.17:

In Liouville's theorem it is importatnt, that the function is entire.

Remark 2.18:

A holomorphic function $f:\Omega\to\mathbb{C}$ can have different power series representation at different points. For example,

$$f(z)=\frac{1}{1-z},\quad \Omega=\mathbb{C}\setminus\{1\}.$$

In $|z|<1$ we have

$$f(z)=\sum_{n=0}^{\infty}z^n.$$

In $|z+1|<2$ we have

$$\frac{1}{1-z}=\frac{1}{2-(z+1)}=\frac{1}{2}\cdot\frac{1}{1-\frac{z+1}{2}}=\sum_{n=0}^{\infty}\frac{(z+1)^n}{2^{n+1}}.$$

Note, that a non-zero holomorphic function can have infinetly many zeroes. For example $f(z)=\cos(z)$. But we will see, that zeroes are isolated, i.e. there exists a neighborhood of a zero, which does not contain any other zeroes.

Definition 2.19: Limit Point

$z_0\in\mathbb{C}$ is a LIMIT POINT of a set Ω, if $\exists$ a sequence $(z_n)_{n\ge1}$ in $\Omega\setminus\{z_0\}$ such that $\lim z_n=z_0$. Hence, $\forall\varepsilon>0$,

$$\Omega\cap(D_\varepsilon(z_0)\setminus\{z_0\})\neq\varnothing.$$

Definition 2.20:

Let Ω be open, $f\in\mathcal{H}(\Omega)$ and $z_0\in\Omega$. The ORDER of f at z_0, denoted by $\operatorname{ord}_{z_0}(f)$, $n_{z_0}(f)$ or $\nu_{z_0}(f)$, is either 0 if $f(z_0)\neq 0$ or the smallest integer k such that

$$f(z_0)=0, f'(z_0)=0,\dots,f^{(k-1)}(z_0)=0, f^{(k)}(z_0)\neq 0.$$

And ∞ if this is does not happen for any k.

Proposition 2.21:

Let Ω be open, $f\in\mathcal{H}(\Omega)$ and $z_0\in\Omega$. Then,

1. If $\operatorname{ord}_{z_0}(f)=\infty$ then $f(z)\equiv 0$ for any $z\in D_r(z_0)$ such that $D_r(z_0)\subset\Omega$.

2. If $\operatorname{ord}_{z_0}(f)\neq\infty$ then $\exists!$ function $h(z)\in\mathcal{H}(D_r(z_0))$ and $n\in\mathbb{Z}, n\ge 0$, such that

$$f(z)=(z-z_0)^nh(z),\quad h(z_0)\neq 0.$$

In this case $n=\operatorname{ord}_{z_0}(f)$.

3. For any $f,g\in\mathcal{H}(\Omega)$, we have

$$\operatorname{ord}_{z_0}(f+g)\ge\min\{\operatorname{ord}_{z_0}(f),\operatorname{ord}_{z_0}(g)\}$$
$$\operatorname{ord}_{z_0}(fg)=\operatorname{ord}_{z_0}(f)+\operatorname{ord}_{z_0}(g).$$

Proof. 1. If $f\in\mathcal{H}(\Omega)$, $z_0\in\Omega$, then we have a power series representation for $D_r(z_0)\subset\Omega$:

$$f(z)=\sum_{n=0}^{\infty}\frac{f^{(n)}(z_0)}{n!}(z-z_0)^n.$$

Since $\operatorname{ord}_{z_0}(f) = \infty$, we have $f^{(n)}(z_0) = 0$ for all $n \geq 0$. Hence, $f(z) \equiv 0$ for any $z \in D_r(z_0)$.

2. If $\operatorname{ord}_{z_0}(f) \neq \infty$, by definition, $\exists k \geq 0$ such that

$$f(z_0) = 0, f'(z_0) = 0, \ldots, f^{(k-1)}(z_0) = 0, f^{(k)}(z_0) \neq 0.$$

Once again, $\exists r > 0$ such that $D_r(z_0) \subset \Omega$ and we have a power series representation

$$f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!}(z - z_0)^n.$$

Since $f^{(n)}(z_0) = 0$ for $n = 0, 1, \ldots, k-1$ and $f^{(k)}(z_0) \neq 0$, we see, that the first $k-1$ terms are zero and the k-th term is non-zero. Hence, we can write

$$f(z) = (z - z_0)^k \underbrace{\sum_{n=0}^{\infty} \frac{f^{(n+k)}(z_0)}{(n+k)!}(z - z_0)^n}_{:= h(z)}.$$

As seen, power series are holomorphic, hence $h(z) \in \mathcal{H}(D_r(z_0))$ and $h(z_0) = \frac{f^{(k)}(z_0)}{k!} \neq 0$.

Note, since $h \in \mathcal{H}(D_r(z_0))$, it is also continuous in $D_r(z_0)$ and $h(z_0) \neq 0$. Thus $\exists r' > 0$ such that $h(z) \neq 0$ in $D_{r'}(z_0)$.

Next, note that k, h are unique if $f(z) = (z-z_0)^k h(z) = (z-z_0)^m g(z)$ with $h(z_0) \neq 0$ and $g(z_0) \neq 0$. If we assume h, g holomorphic, and w.l.o.g. $m > k$, then we have

$$f(z) = (z - z_0)^k (z - z_0)^{m-k} g(z) = (z - z_0)^k h(z).$$

If $z \neq z_0$, then

$$h(z) = (z - z_0)^{m-k} g(z).$$

Lec 10

But then, for $z = z_0$, we get $h(z_0) = 0$ unless $m - n = 0$ implying $m = n$. □

Theorem 2.22: Isolated Zeros Theorem

Let $\Omega \subset \mathbb{C}$, $f \in \mathcal{H}(\Omega)$, $z_0 \in \Omega$. Assume $f(z_0) = 0$ so $\operatorname{ord}_{z_0} f \geq 1$. If $\operatorname{ord}_{z_0} f \neq \infty$, then $\exists \delta > 0$ such that $f(z) \neq 0, \forall z \in D_\delta(z_0) \setminus \{z_0\}$.

Proof. By the previous proposition 2. $\exists h, n$ such that

$$f(z) = (z - z_0)^n h(z), \quad h \in \mathcal{H}(D_\delta(z_0)), h(z_0) \neq 0.$$

Since h is holomorphic, it is continuous. Hence $\exists \delta < r$ such that $h(z) \neq 0, \forall z \in D_\delta(z_0)$. Therefore, $f(z) \neq 0, \forall z \in D_\delta(z_0) \setminus \{z_0\}$. □

Denote by $D^*_\delta(z_0) = D_\delta(z_0) \setminus \{z_0\}$ the punctured disk of radius δ centered at z_0.

Theorem 2.23:

Let Ω open and connected. Let $f \in \mathcal{H}(\Omega)$. Then the following are equivalent:

a. $f \equiv 0$ on Ω.

b. $\exists a \in \Omega$, such that $f^{(n)}(a) = 0, \forall n \in \mathbb{Z}_{\geq 0}$.

c. $\{z \in \Omega | f(z) = 0\}$ has a limit point in Ω.

Proof. Clearly $a \Rightarrow c$ since then the set $\{z \in \Omega | f(z) = 0\} = \Omega$. We'll show that $c \Rightarrow b \Rightarrow a$.

$c \Rightarrow b$: Let $\mathcal{Z} = \{z \in \Omega | f(z) = 0\}$. By assumption, $\exists a \in \Omega$ which is a limit point of $\mathcal{Z}$. Let $r > 0$, such that $D_r(a) \subset \Omega$. Since f is holomorphic, it is continuous. Since a is a limit point, hence $\exists z_n \in \mathbb{Z} \setminus \{a\}$ such that $z_n \to a$. By continuity, $f(z_n) \to f(a)$. But $f(z_n) = 0$, hence $f(a) = 0$.

We claim, that $f^{(n)}(a) = 0, \forall n \in \mathbb{Z}_{\geq 0}$. Suppose not, so $\exists n > 0$ such that

$$f(a) = 0 = \cdots = f^{(n-1)}(a), \quad f^{(n)}(a) \neq 0.$$

Then as in the proof of the proposition, since f is holomorphic in $D_r(a) \subset \Omega$, it has a power series

$$f(z) = \sum_{k=0}^{\infty} a_k (z - a)^k, \quad z \in D_r(a).$$

Thus $f(z) = (z - a)^n g(z)$ with $g(a) \neq 0$ and g holomorphic. Since g is continuous, $\exists \varepsilon > 0, D_\varepsilon(a) \subset D_r(a)$ such that $g(z) \neq 0, \forall z \in D_\varepsilon(a)$. Hence $f(z) \neq 0, \forall z \in D^*_\varepsilon(a)$. Hence $\mathcal{Z} \cap D^*_\varepsilon(a) = \varnothing$, which contradicts the fact that a is a limit point of $\mathcal{Z}$. Hence $f^{(n)}(a) = 0, \forall n \in \mathbb{Z}_{\geq 0}$.

$b \Rightarrow a$: Let $A = \{z \in \Omega \mid f^{(n)}(z) = 0, \forall n \geq 0\}$. By assumption, $\exists$ at least one point $a \in A$, hence $A \neq \varnothing$. Since Ω is connected, if we can show that A is clopen, then $A = \Omega$ and hence $f \equiv 0$ on Ω.

A is open: Let $c \in A$. Let $r > 0$ such that $D_r(c) \subset \Omega$. Then

$$\frac{f}{z} = \sum_{n=0}^{\infty} a_n (z - c)^n, \quad z \in D_r(c).$$

But recall that $a_n = \frac{f^{(n)}(c)}{n!} = 0, \forall n \geq 0$. Hence $f(z) = 0, \forall z \in D_r(c)$. Thus, $D_r(c) \subset A$, hence A is open.

A is closed: We want to show that if $\{z_k\}$ is a sequence in A with $z_k \to c \in \Omega$, then $c \in A$. Let $c \in \Omega$ be a limit point of $\{z_k\} \subset A$. By definition,

$$f^{(n)}(z_k) = 0 \quad \forall n \geq 0, \forall k \in \mathbb{Z}_{>0}.$$

Since f is holomorphic, all its derivatives are continuous. So

$$0 = \lim f^{(n)}(z_k) = f^{(n)}(\lim z_k) = f^{(n)}(c) \quad \forall n \geq 0.$$

Thus $c \in A$, hence A is closed. Since Ω is connected, $A = \Omega$ and hence $f \equiv 0$ on Ω. □

From this, the identity theorem follows immediately.

Theorem 2.24: Identity Theorem

Let Ω be open and connected. Let $f \in \mathcal{H}(\Omega)$ let $\mathscr{Z}$ be an infinite set with a limit point $z_0 \in \Omega$. If $f(z) = 0 \forall z \in \mathscr{Z}$, then $f \equiv 0$.

Corollary 2.25:

Suppose $f, g \in \mathcal{H}(\Omega)$ where Ω is open and connected. If $f(z) = g(z)$ for all z in an nonempty open subset $U \subset \Omega$ (or more generally for a set $\mathscr{Z}$ with a limit point in Ω), then $f \equiv g \ \forall z \in \Omega$.

Proof. Apply the identity theorem to $f - g$. □

Remark 2.26:

Note that Ω connected is essential. e.g. let $\Omega = U_1 \cup U_2$ where U_i are open and disjoint. Let $f(z) = \chi_{U_1}(z)$ and $g(z) = 0$, then $f \equiv g$ on U_2 but $f \not\equiv g$ on Ω.

Furthermore, the condition that the limit point $z_0 \in \Omega$ is also essential. e.g. Let $f : \Omega := \mathbb{C} \setminus \{0\} \to \mathbb{C}$ be defined by $z \mapsto \sin(\frac{\pi}{z})$. Clearly $f \not\equiv 0$ on Ω. However, looking at $f(\frac{1}{n}) = 0$ for all $n \in \mathbb{Z} \setminus \{0\}$, we see that f vanishes on an infinite set with a limit point at $0 \notin \Omega$.

Lastly, the theorem is sometimes referred as the principle of analytic continuation. If f is holomorphic on Ω and $\Omega \subset \tilde{\Omega}$ where $\tilde{\Omega}$ is open and connected, then there exists at most one holomorphic function $\tilde{f}$ on $\tilde{\Omega}$ such that $\tilde{f}|_\Omega = f$. If such a $\tilde{f}$ exists, we say that f has an ANALYTIC CONTINUATION to $\tilde{\Omega}$.

The identity theorem makes it clear, that sin, cos, exp : $\mathbb{R} \to \mathbb{R}$ can be extended *uniquely* to $\mathbb{C}$. The functional equations can also be extended. For example, $\sin z, \cos z$ are entire and on the real line $\sin^2 x + \cos^2 x = 1$. This implies that

$$\sin^2 z + \cos^2 z = 1, \quad \forall z \in \mathbb{C}.$$

This is because $f(z) = \sin^2 z + \cos^2 z = 1 = g(z)$ for all $z \in \mathbb{R}$, and thus by identity theorem the result follows. Similarly, we get from $\exp(x+y) = \exp(x)\exp(y)$ for all $x, y \in \mathbb{R}$ that

$$\exp(z+w) = \exp(z)\exp(w), \quad \forall z, w \in \mathbb{C}.$$

Theorem 2.27:

Let $f, g \in \mathcal{H}(\Omega)$, in an open connected set. If $f \cdot g = 0$ on Ω, then either $f \equiv 0$ or $g \equiv 0$ on Ω.

Proof. Suppose $f \not\equiv 0$. We want to show that $g \equiv 0$. Since $f \not\equiv 0$, $\exists z_0 \in \Omega$ such that $f(z_0) \neq 0$. By continuity, $\exists \delta > 0$ such that $f(z) \neq 0, \forall z \in D_\delta(z_0)$.

Since $f(z)g(z) = 0 \forall z \in \Omega$, we have $g(z) = 0, \forall z \in D_\delta(z_0)$. Hence g vanishes on an open set, and by the identity theorem, $g \equiv 0$ on Ω. □

Remark 2.28:

The theorem is not true for real functions. Even for C^∞ functions. One needs the stronger condition of analyticity.

Remark 2.29:

Holomorphic functions on a non-empty subset $\Omega \subset \mathbb{C}$ form a commutative ring with unity. Since the sum and product of holomorphic functions are holomorphic, the above theorem says that if Ω is also connected, then they form a ring with no zero divisors.

Example 2.30:

Take $f(z) = \sum_{n=0}^{\infty} z^n$ converges and hence defines a holomorphic function in $D_1(0) = \Omega$. Take $\tilde{\Omega} = \mathbb{C} \setminus \{1\}$. Then $\Omega \subset \tilde{\Omega}$ and we can define $\tilde{f}(z) = \frac{1}{1-z}$ which is holomorphic on $\tilde{\Omega}$. Clearly $\tilde{f}|_\Omega = f$. Hence $\tilde{f}$ is an analytic continuation of f to $\tilde{\Omega}$.

Remark 2.31:

Not every function can be extended analytically to a larger set. For example, take

$$\sum_{n=0}^{\infty} z^{n!}.$$

If $|z| < 1$, then the series converges and hence defines a holomorphic function on $D_1(0)$.

Lec 11

We want to look at a converse theorem of Goursat's theorem.

Theorem 2.32: Morera's theorem

Let $\Omega \subset \mathbb{C}$ be an open set, $f : \Omega \to \mathbb{C}$ continuous. Assume that for any open disc $D \subset \Omega$ and any triangle T whose interior is also contained in D, we have

$$\int_T f(z)\, dz = 0.$$

Then f is holomorphic on Ω.

Proof. Let $z_0 \in \Omega$ and $D_r(z_0) \subset \Omega$. For $z \in D_r(z_0)$, we define

$$F(z) := \int_{[z_0, z]} f(w)\, dw,$$

where $[z_0, z]$ is the line segment from z_0 to z.

Then, for small h, such that $z + h \in D_r(z_0)$, we have by assumption

$$F(z+h) - F(z) = \int_{[z, z+h]} f(w)\, dw.$$

From continuity of f, it follows that

$$\lim_{h \to 0} \frac{F(z+h) - F(z)}{h} = f(z).$$

This will show that F is holomorphic and $F' = f$. Since F is holomorphic, it follows that f is holomorphic as well.

It remains to show the continuity argument. We have

$$\begin{aligned} F(z+h) - F(z) &= \int_{[z, z+h]} f(w) - f(z) + f(z)\, dw \\ &= f(z) \int_{[z, z+h]} dw + \int_{[z, z+h]} f(w) - f(z)\, dw \\ &= f(z)h + \int_{[z, z+h]} f(w) - f(z)\, dw. \end{aligned}$$

However we can bound the second integral by

$$\left| \int_{[z, z+h]} f(w) - f(z)\, dw \right| \leq |h| \sup_{w \in [z, z+h]} |f(w) - f(z)|.$$

So

$$\left| \frac{F(z+h) - F(z)}{h} - f(z) \right| \leq \sup_{w \in [z, z+h]} |f(w) - f(z)|.$$

Letting $h \to 0$, since f is continuous, we have that the right hand side goes to 0, which proves the claim. □

2.1 Sequences of holomorphic functions

As a quick reminder: In real analysis if we have point wise convergence of a sequence of continuous functions, the limit function is not necessarily continuous.

Example 2.33: Pointwise convergence of continuous functions

Let $f_n : [0,1] \to \mathbb{R}, f_n(x) = x^n$. Then $f_n \to f$ pointwise, where

$$f(x) = \begin{cases} 0 & x \in [0,1) \\ 1 & x = 1 \end{cases}.$$

This can be repaired by looking at a stronger notion of convergence, called uniform convergence. Similarly, uniform limits of integrable functions are integrable. However, uniform convergence of differentiable functions is not necessarily differentiable.

Example 2.34: Uniform convergence of differentiable functions

Let $f_n : [-1,1] \to \mathbb{R}, f_n(x) = \sqrt{\frac{1+nx^2}{n}}$. Then $f_n \to f$ uniformly, but f is not differentiable at 0. In fact, $f(x) = |x|$.

Fortunately for holomorphic functions, uniform convergence preserves holomorphicity.

Definition 2.35: Locally uniformly convergent sequence of functions

Let $\Omega \subset \mathbb{C}$ be open, $f_n : \Omega - \mathbb{C}$ a sequence of functions. Then $(f_n)_{n\geq 1}$ is called LOCALLY UNIFORMLY CONVERGENT or UNIFORMLY CONVERGENT ON COMPACTA if the following equivalent conditions hold:

1. $\forall a \in \Omega \exists \varepsilon > 0$ such that $D_\varepsilon(a) \subset \Omega$ such that $f_n \to f$ uniformly on $D_\varepsilon(a)$.

2. For every compact set $K \subset \Omega$, $f_n \to f$ uniformly on K.

Proposition 2.36: Continuity of loc. unif. convergent functions

Let $(f_n)_{n\geq 1}$ be a sequence of continuous functions $f_n : \Omega \to \mathbb{C}$, which converges locally uniformly to $f : \Omega \to \mathbb{C}$. Then f is continuous.

Proof. Exercise □

Theorem 2.37: Weierstrass convergence theorem

Let $(f_n)_{n\geq 1}$ be a sequence of holomorphic functions on $\Omega \subset \mathbb{C}$ open. If f_n converges locally uniformly to $f : \Omega \to \mathbb{C}$, then f is holomorphic on Ω.

Proof. Since f_n are holomorphic, they are also continuous. Hence, by Proposition 2.36 , f is also continuous. To prove that f is holomorphic, we will use Morera's theorem [2.32].

Let $T \subset D \subset \Omega$ with T a triangle and D an open disc such that the interior of T is contained in D. Since f_n are holomorphic, by Goursat's theorem,

$$\int_T f_n(w)\, dw = 0.$$

Since T is a triangle, it is compact. So by assumption $f_n \to f$ uniformly on T. Thus we can bound

$$\left|\int_T f_n(w)\, dw - \int_T f(w)\, dw\right| = \left|\int_T f_n(w) - f(w)\, dw\right| \leq \sup_{w\in T} |f_n(w) - f(w)| \cdot \text{length}(T).$$

Letting $n \to \infty$, the right hand side goes to 0, by definition of uniform convergence.

By Morera's theorem [2.32], it follows that f is holomorphic on Ω. □

The theorem above can be generalized as follows

Theorem 2.38:

Given $(f_n)_{n\geq 1}$ a sequence of holomorphic functions on $\Omega \subset \mathbb{C}$ open, such that f_n converges locally uniformly to $f : \Omega \to \mathbb{C}$, then $f_n' \to f'$ locally uniformly on Ω.

Lec 12

Proof. Let $z_0 \in \Omega, r > 0$ such that $\overline{D(z_0, r)} \subset \Omega$. Then $f_n \to f$ uniformly on $\overline{D(z_0, r)}$. Let $s > r$ such that $\overline{D(z_0, s)} \subset \Omega$. Let $\sigma = \frac{r+s}{2}$. By Cauchy's integral formula, we have for any $z \in D(z_0, r)$,

$$f'(z) = \frac{1}{2\pi i} \int_{|w-z_0|=\sigma} \frac{f(w)}{(w-z)^2} d\dot{w}.$$

Similarly, $\forall z \in \overline{D_r(z_0)}$, we have

$$f_n'(z) = \frac{1}{2\pi i} \int_{|w-z_0|=\sigma} \frac{f_n(w)}{(w-z)^2} d\dot{w}.$$

For such z by uniform convergence, we have

$$|f_n'(z) - f'(z)| = \left|\frac{1}{2\pi i} \int_{|w-z_0|=\sigma} \frac{f_n(w) - f(w)}{(w-z)^2} d\dot{w}\right| \leq \frac{1}{2\pi} 2\pi\sigma \sup_{|w-z_0|=\sigma} \left|\frac{f_n(w) - f(w)}{(w-z)^2}\right|.$$

For $w \in C_\sigma(z_0)$ and $z \in \overline{D_r(z_0)}$, we have

$$|w - z_0| = \sigma, \quad |z - z_0| \leq r$$
$$\Rightarrow |w - z| \geq |w - z_0| - |z - z_0| \geq \sigma - r.$$

Thus, we get

$$|f_n'(z) - f'(z)| \leq \frac{\sigma}{(\sigma - r)^2} \sup_{|w-z_0|=\sigma} |f_n(w) - f(w)| \to 0 \quad \text{as } n \to \infty.$$

Where we used that uniform convergence implies convergence in the supremum norm. □

Remark 2.39:

These theorems are used to prove holomorphicity of functions defined by infinite sums of holomorphic functions $F(z) := \sum_{k=1}^\infty f_k(z)$, where f_k are holomorphic functions.

Usually, this is done by taking the sequence of partial sums $S_N := \sum_{k=1}^N f_k(z)$ and showing that S_N converges uniformly to F on compact subsets of Ω.

A useful criteria for uniform convergence of a series of functions is the Weierstrass M-test.

Theorem 2.40: Weierstrass M-test

Let $(f_n)_{n\geq 1} : \Omega \to \mathbb{C}$ be a sequence of holomorphic functions, $U \subset \Omega$ open, non-empty. Suppose there exists a sequence of *real* numbers $M_n \geq 0$ such that $|f_n(z)| \leq M_n$ for all $z \in U$ and $\sum_{n=1}^\infty M_n < \infty$. Then the series $\sum_{n=1}^\infty f_n(z)$ converges absolutely and uniformly on U.

Proof. Exercise. □

Example 2.41: Riemann zeta function

Let $s \in \mathbb{C}$, $s = \sigma + it$ with $\sigma, t \in \mathbb{R}, n \in \mathbb{N}$. The function

$$s \mapsto n^s = \exp(s \log n)$$

is entire. Furthermore,

$$|n^s| = |\exp(s \log n)| = |\exp(\sigma \log n + it \log n)| = |\exp(\sigma \log n)| = n^\sigma.$$

Then the series

$$\zeta(s) = \sum_{n=1}^\infty \frac{1}{n^s}$$

converges absolutely and uniformly on every half plane $U_\delta = \{s \in \mathbb{C} : \Re(s) > 1 + \delta\}$ for $\delta > 0$ and is holomorphic in $\{s \in \mathbb{C} \mid \Re(z) > 1\}$.

Solution. For each $\delta > 0$, if $\Re(s) = \sigma \geq 1 + \delta > 1$, then the series

$$\sum_{n=1}^\infty \frac{1}{n^s},$$

is uniformly bounded by the convergent series

$$\sum_{n=1}^{\infty} \frac{1}{n^{1+\delta}} < \infty.$$

Hence,

$$|\zeta(s)| \leq \sum_{n=1}^{\infty} \frac{1}{|n^s|} \leq \sum_{n=1}^{\infty} \frac{1}{n^{1+\sigma}}.$$

Hence, $\sum_{n=1}^{\infty} \frac{1}{n^s}$ converges absolutely and uniformly on every half plane $\Re(s) > 1+\delta$ for $\delta > 0$. Hence it defines a holomorphic function in $\{s \in \mathbb{C} \mid \Re(s) > 1\}$, because every compact subset of $\{s \in \mathbb{C} \mid \Re(s) > 1\}$ is contained in some half plane $\Re(s) > 1 + \delta$ for some $\delta > 0$.

Example 2.42: **Theta function**

For $z \in \mathbb{H} = \{z = x + iy \in \mathbb{C} \mid y > 0\}$, the upper half plane, define the theta function

$$\theta(z) := \sum_{n=-\infty}^{\infty} e^{2\pi i n^2 z} = 1 + 2\sum_{n=1}^{\infty} e^{2\pi i n^2 z}.$$

Proposition 2.43:

$\forall z \in \mathbb{H}$, the series $\theta(z)$ converges and defines a holomorphic function on $\mathbb{H}$.

Proof. We'll prove that it converges uniformly on any subset of the form

$$\mathbb{H}_\delta = \{z = x + iy \in \mathbb{C} \mid y > \delta\}, \quad \delta > 0.$$

Since any compact subset of $\mathbb{H}$ is contained in such a set $\mathbb{H}_\delta$ for some $\delta > 0$, this will imply that $\theta(z)$ is holomorphic on $\mathbb{H}$.

For $z \in \mathbb{H}_\delta$, we have

$$\begin{aligned} \left|e^{2\pi i n^2 z}\right| &= \left|e^{2\pi i n^2 (x+iy)}\right| \\ &= \left|e^{2\pi i n^2 x} e^{-2\pi n^2 y}\right| \\ &= e^{-2\pi n^2 y} \\ &\leq e^{-2\pi n y}. \end{aligned}$$

Since $y \geq \delta$, we have

$$e^{-2\pi n y} \leq e^{-2\pi n \delta} < 1.$$

Hence,

$$\left|\sum_{n=1}^{\infty} e^{2\pi i n^2 z}\right| \leq \sum_{n=1}^{\infty} e^{-2\pi n \delta} < \infty.$$

The Weierstrass M-test gives us uniform convergence in $\mathbb{H}_\delta$ and hence the theorem on local uniform convergence gives the holomorphicity. □

Remark 2.44:

There is actually a relation between $\zeta(s)$ and $\theta(z)$. This relation is behind the analytic continuation of $\zeta(s)$ to the region $\mathbb{C} \setminus \{1\}$.

Remark 2.45:

$$\pi^{-s/2}\Gamma(s/2)\zeta(s) = \frac{1}{2}\int_0^{\infty} (\theta(it) - 1)t^s \frac{dt}{t}.$$

The Lefthandside is defined for $\Re(s) > 1$ while the righthandside makes sense for $\mathbb{C}$
$\{0, 1\}$. This gives an analytic continuation of $\zeta(s)$ to $\mathbb{C} \setminus \{1\}$.

Next, we look at functions defined by integrals and ask when they are holomorphic. Many special functions are of the form

$$f(z) := \int_a^b F(z,t)dt.$$

Example 2.46:

The gamma function is defined by

$$\Gamma(z) := \lim_{\varepsilon \to \infty} \int_{\frac{1}{\varepsilon}}^{\varepsilon} e^{-t} t^z \frac{\mathrm{d}t}{t}.$$

Theorem 2.47:

Let $\Omega \subset \mathbb{C}$ be open, $I = [a, b] \subset \mathbb{R}$. Let $F : \Omega \times I \to \mathbb{C}$ be a function with the following properties:

1. F is continuous on $\Omega \times I$.
2. For each $t_0 \in I$, the function $f_{t_0}(z) := F(z, t_0)$ is holomorphic on Ω.

Then the function $f : \Omega \to \mathbb{C}$ defined by

$$f(z) := \int_a^b F(z,t)dt.$$

is holomorphic on Ω.

Proof. The idea is to use Riemann sums to approximate the integral. Let

$$f_n(z) := \frac{b-a}{n} \sum_{j=0}^{n-1} F(z, a + \frac{b-a}{n} j).$$

Then $f_n(z)$ is a finite sum of holomorphic functions and hence holomorphic. We want to show, that $f_n \to f$ uniformly on compact subsets of Ω.

Let $K \subset \Omega$ be compact. Since F is continuous on the compact set $K \times I$, it is uniformly continuous. Hence, $\forall \varepsilon > 0, \exists \delta > 0$ such that $\forall (z_i, t_i) \in K \times I, i = 1, 2$ with $|(z_1, t_1) - (z_2, t_2)| < \delta$, we have

$$|F(z_1, t_1) - F(z_2, t_2)| < \frac{\varepsilon}{b-a}.$$

Let n be large enough such that $\frac{b-a}{n} < \delta$. Then, $\forall z \in K$, we have

$$\begin{aligned} |f_n(z) - f(z)| &= \sum_{j=0}^{n-1} \int_{a+j\frac{b-a}{n}}^{a+(j+1)\frac{b-a}{n}} |F(z, a + \frac{b-a}{n} j) - F(z,t)|dt \\ &\leq \sum_{j=0}^{n-1} \int_{a+j\frac{b-a}{n}}^{a+(j+1)\frac{b-a}{n}} \frac{\varepsilon}{b-a} dt \\ &= \sum_{j=0}^{n-1} \frac{b-a}{n} \frac{\varepsilon}{b-a} = n\frac{b-a}{n}\frac{\varepsilon}{b-a} = \varepsilon \end{aligned}$$

Since this holds $\forall z \in K$, we have uniform convergence on K. Since K was arbitrary, we have uniform convergence on compact subsets of Ω. Hence, by the theorem on local uniform convergence, f is holomorphic on Ω. □

Example 2.48: **J Bessel function**

Many special functions that arise as solutions of differential equations also have integral representations. For example the J bessel function $J_n(z)$ is defined as the solution of the differential equation

$$z^2\frac{\mathrm{d}^2 f}{\mathrm{d}z^2} + z\frac{\mathrm{d}f}{\mathrm{d}z} + (z^2 - n^2)f = 0.$$

For $n \in \mathbb{Z}$, it has the integral representation

$$J_n(z) = \frac{1}{2\pi}\int_{-\pi}^{\pi} e^{iz\sin t}e^{-int}dt.$$

Remark 2.49:

With some extra work, one can also show that $f'(z)$ is given by

$$f'(z) = \int_a^b \frac{\partial F}{\partial z}(z,t)dt.$$

In other words, one can interchange the order of differentiation and integration.

3 Meromorphic Functions and the Logarithm

Lec 13

We now like to extend Cauchy's theorem to functions with singularities. As we will see, there exist three types of singularities. We have poles as well as removable, and essential singularities.

To describe these singularities, we first need to define what an isolated singularity is.

Definition 3.1: **Isolated Singularity**

Let $z_0 \in \mathbb{C}$. z_0 is called a (possible) ISOLATED SINGULARITY of f if $\exists r > 0$ such that f is holomorphic in the punctured disk $D_r^*(z_0)$.

The easiest type of singularity is a removable singularity. This is a singularity that can be "removed" by defining the function at the singularity in a way that makes it holomorphic.

Definition 3.2: **Removable Singularity**

An isolated singularity z_0 of a function $f \in \mathcal{H}(\Omega \setminus \{z_0\})$ is called REMOVABLE if f is holomorphically extendable to z_0, i.e. $\exists F : \Omega \to \mathbb{C}$ which is holomorphic and $\forall z \in \Omega \setminus \{z_0\}$, $F(z) = f(z)$.

Theorem 3.3:

Let $z_0 \in \mathbb{C}$. Then the following assertions for a function $f \in \mathcal{H}(\Omega \setminus \{z_0\})$ are equivalent:

1. f is holomorphically extendable to Ω.
2. f is continuouslly extendable to Ω.
3. f is bounded in a neighborhood of z_0.
4. $\lim\limits_{z\to z_0}(z - z_0)f(z) = 0$.

Proof. Exercise 5.4 □

A corollary of this is Riemann's theorem on removable singularities.

Theorem 3.4: **Riemann's Theorem on Removable Singularities**

Suppose f is holomorphic on an open set Ω except possibly at a point $z_0 \in \Omega$. If f is bounded in $D_r^*(z_0)$ for some $r > 0$, then z_0 is a removable singularity of f.

Example 3.5:

The function $f(z) = \frac{\sin z}{z}$ has a removable singularity at $z = 0$ since

$$\lim_{z\to 0} f(z) = \lim_{z\to 0}\frac{\sin z}{z} = 1.$$

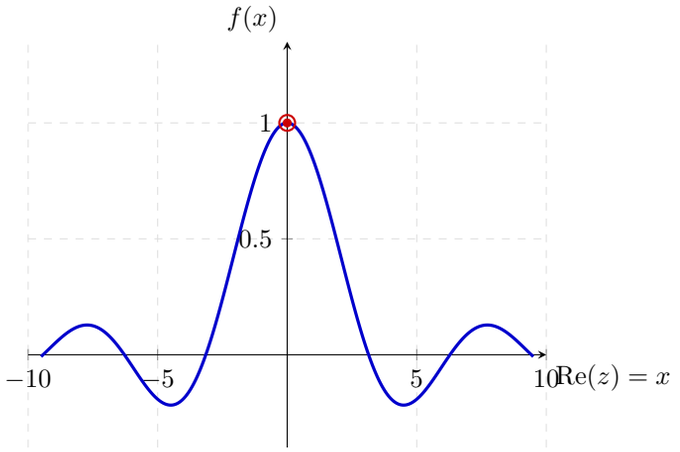


Figure 8: The function $f(z) = \frac{\sin z}{z}$ has a removable singularity at $z = 0$. The limit $\lim_{z\to 0} f(z) = 1$ exists, and the function can be extended to be holomorphic at $z = 0$ by defining $f(0) = 1$.

Corollary 3.6:

Let f be a function with an isolated singularity at z_0. Then the singularity is removable if and only if f is bounded in a punctured disk around z_0 if and only if $\lim_{z\to z_0} f(z)$ exists.

Example 3.7:

The function $f(z) = \frac{z}{e^z - 1}$ has a removable singularity at $z = 0$ since

$$\lim_{z\to 0} f(z) = \lim_{z\to 0} \frac{z}{e^z - 1} = \lim_{z\to 0} \frac{1}{e^z} = 1.$$

If f does not have a removable singularity at z_0, the f is not bounded near z_0. The question is now if this unboundedness is similar to the unboundedness of $\frac{1}{z-z_0}$ or $\frac{1}{(z-z_0)^n}$ for some $n \in \mathbb{N}$. If such an n exists, $(z - z_0)^n f(z)$ is bounded. This leads us to the definition of poles.

Definition 3.8: Pole

Let $f \in \mathcal{H}(\Omega \setminus \{z_0\})$. Suppose $\exists n \geq 0$ such that $(z - z_0)^n f(z)$ is bounded in a neighborhood of z_0. Then z_0 is called a POLE of f. The natural number

$$m := \min\{n \in \mathbb{N}_0 : (z-z_0)^n f(z) \text{ bdd in nbhd of } z_0\},$$

is called the ORDER of the pole z_0. If $m = 1$, we say that z_0 is a SIMPLE POLE.

If z_0 is neither a pole nor a removable singularity, we call it an essential singularity.

Definition 3.9: Essential Singularity

An isolated singularity z_0 of $f \in \mathcal{H}(\Omega \setminus \{z_0\})$ is called an ESSENTIAL SINGULARITY if it is neither a removable singularity nor a pole.

Remark 3.10:

Complex functions can have non-isolated singularities. For example, $f(z) = \tan(\frac{1}{z})$ has singularities at $z = \frac{1}{n\pi}$ for all $n \in \mathbb{Z} \setminus \{0\}$, which accumulate at $z = 0$. However, in this course, we will only consider isolated singularities.

Recall the theorem on zeroes of holomorphic functions. It stated that if $f \not\equiv$ is holomorphic in a connected open set Ω and has a zero at $z_0 \in \Omega$, then $\exists r > 0$ and a unique non-vanishing function, which is holomorphic in $D_r(z_0)$, such that $f(z) = (z - z_0)^m g(z)$ for all $z \in D_r(z_0)$. Where n is the order of the zero.

Lec 14

Theorem 3.11: Different Notions of Poles

For $m \in \mathbb{N}$, $m \geq 1$, the following statements for a function f which is Holomorphic except at z_0 are equivalent:

1. f has a pole of order m at z_0.
2. $\exists r > 0$, $g \in \mathcal{H}(D_r(z_0))$ such that $g(z_0) \neq 0$ and $f(z) = \frac{g(z)}{(z-z_0)^m}$ for all $z \in D_r^*(z_0)$.
3. $\exists r > 0$, such that $D_r(z_0) \subset \Omega$ and $h \in \mathcal{H}(D_r(z_0))$ such that $h(z_0) \neq 0 \forall z \in D_r^*(z_0)$, and $f(z) = \frac{1}{h(z)}$.

Proof. $1 \Rightarrow 2$: f has a pole of order m at z_0, such that m is minimal. By Riemann's theorem on removable singularities, $\exists g \in \mathcal{H}(D_r(z_0))$ such that $g(z_0) \neq 0$ and $g(z) = (z - z_0)^m f(z)$ for all $z \in D_r^*(z_0)$. Thus, $f(z) = \frac{g(z)}{(z-z_0)^m}$ for all $z \in D_r^*(z_0)$.

If $g(z_0) = 0$, then we have $g(z) = (z - z_0)h(z)$ where h is holomorphic in $D_r(z_0)$ in particular this means h is bounded implying the pole is of order $m - 1$ which is a contradiction. Thus, $g(z_0) \neq 0$.

$2 \Rightarrow 3$: Suppose $\exists g \in H(D_r(z_0))$ such that $g(z_0) \neq 0$ and $f(z) = \frac{g(z)}{(z-z_0)^m}$ for all $z \in D_r^*(z_0)$. Let $h(z) = \frac{(z-z_0)^m}{g(z)}$ for all $z \in D_r(z_0)$. Since g is continuous, there is a disk around z_0 such that $g(z) \neq 0$ for all $z \in D_r(z_0)$.

Then, $h(z) \neq 0$ except for z_0 and $h \in \mathcal{H}(D_r(z_0))$. Thus, $f(z) = \frac{1}{h(z)}$ for all $z \in D_r^*(z_0)$. Note, that h has a zero of order m at z_0, since $g(z_0) \neq 0$.

$3 \Rightarrow 1$: Suppose $\exists r > 0, h \in \mathcal{H}(D_r(z_0))$ such that $h(z) \neq 0$ for all $z \in D_r^*(z_0)$ and $f(z) = \frac{1}{h(z)}$ for all $z \in D_r^*(z_0)$. h has a zero of order m, we can write $h(z) = (z - z_0)^m g(z)$ with $g \in \mathcal{H}(D_s(z_0))$ such that $g(z) \neq 0$ for all $z \in D_s(z_0)$.

Thus, $\frac{1}{g}$ is holomorphic in $D_s(z_0)$ and $f(z) = \frac{1}{(z-z_0)^m g(z)}$ for all $z \in D_s^*(z_0)$. Therefore, f has a pole of order m at z_0. □

Remark 3.12:

f has a pole at z_0 is equivalent to $\lim_{z\to z_0} |f(z)| = \infty$.

Example 3.13:

Let $f(z) = \frac{1}{e^z - 1}$. Then f has a pole of order 1 at $z_0 = 0$ since $e^z - 1$ has a zero of order 1 at $z_0 = 0$.

Example 3.14:

Consider $f(z) = \frac{z}{z^2-1}$. Then $f = \frac{z}{(z-1)(z+1)}$ has a pole of order 1 at $z_0 = 1$ and $z_0 = -1$ since $(z - 1)f(z) = \frac{z}{z+1}$ and $(z+1)f(z) = \frac{z}{z-1}$ are holomorphic in $D_r(1)$ and $D_r(-1)$ respectively for some $1 > r > 0$.

We now want to look at the analogue of the theorem that if f is holomorphic in some neighborhood of z_0 then there is a power series expansion of f around z_0.

Theorem 3.15:

If f has a pole of order n at z_0, then

$$f(z) = \frac{a_{-n}}{(z-z_0)^n} + \frac{a_{-(n-1)}}{(z-z_0)^{n-1}} + \cdots + \frac{a_{-1}}{(z-z_0)} + G(z),$$

for $z \in D_r^*(z_0)$, where $G \in \mathcal{H}(D_r(z_0))$.

Proof. If f has a pole of order m, by the previous theorem we have

$$f(z) = (z-z_0)^{-n} g(z).$$

We can expand g in a power series around z_0 as

$$g(z) = a_0 + a_1(z-z_0) + a_2(z-z_0)^2 + \cdots.$$

But then, we have

$$f(z) = \frac{a_0}{(z-z_0)^n} + \frac{a_1}{(z-z_0)^{n-1}} + \cdots + \frac{a_{n-1}}{(z-z_0)} + G(z).$$

□

Remark 3.16:

The first n terms of the expansion are called the PRINCIPAL PART of f at z_0. We sometimes write

$$P_f(z, z_0) = P_{z_0}(z) = \frac{a_{-n}}{(z-z_0)^n} + \cdots + \frac{a_{-1}}{(z-z_0)}.$$

The coefficient a_{-1}, denoted by $\operatorname{Res}_{z_0}(f)$ is called the RESIDUE of f at z_0.

Remark 3.17:

If f has a simple pole at z_0, then

$$\operatorname{Res}_{z_0}(f) = \lim_{z \to z_0} (z-z_0) f(z).$$

Theorem 3.18:

If f has a pole of order n at z_0, then

$$\operatorname{Res}_{z_0}(f) = \lim_{z \to z_0} \frac{1}{(n-1)!} \frac{d^{n-1}}{dz^{n-1}} \left((z-z_0)^n f(z) \right).$$

Proof. We have $f(z) = P_f(z) + G(z)$, where $G \in \mathcal{H}(D_r(z_0))$. Then, we have

$$(z-z_0)^n f = a_{-n} + \cdots + a_{-1}(z-z_0)^{n-1} + (z-z_0)^n G(z).$$

Differentiating $n-1$ times, we have

$$\frac{d^{n-1}}{dz^{n-1}} \left((z-z_0)^n f(z) \right) = (n-1)! a_{-1} + (z-z_0) H(z).$$

Letting $z \to z_0$, we have

$$\lim_{z \to z_0} \frac{d^{n-1}}{dz^{n-1}} \left((z-z_0)^n f(z) \right) = (n-1)! a_{-1}.$$

□

Example 3.19:

Let $f = \frac{1}{z^2+1}$. Then f has a simple pole at $z_0 = i$ and $z_0 = -i$. We can write $f(z) = \frac{a}{z-i} + \frac{b}{z+i}$ for some $a, b \in \mathbb{C}$. Then, we have

$$\operatorname{Res}_i(f) = \lim_{z \to i} (z-i) f(z) \lim_{z \to i} \frac{1}{z+i} = \frac{1}{2i}.$$

Similarly, we have

$$\operatorname{Res}_{-i}(f) = \lim_{z \to -i} (z+i) f(z) = \lim_{z \to -i} \frac{1}{z-i} = -\frac{1}{2i}.$$

Example 3.20:

The function $f = \frac{1}{(z^2+1)^2}$ has the same poles as the previous example, but they are of order 2. We can write

$$\begin{aligned} \operatorname{Res}_i(f) &= \lim_{z \to i} \frac{d}{dz} \left((z-i)^2 f(z) \right) \\ &= \lim_{z \to i} \frac{d}{dz} \left(\frac{1}{(z+i)^2} \right) = \frac{1}{4i}. \end{aligned}$$

Lemma 3.21:

If f, g are holomorphic at z_0, $f(z_0) \neq 0$ and $g(z)$ has a simple zero at z_0, then $\frac{f(z)}{g(z)}$ has a simple pole at z_0 and

$$\operatorname{Res}_{z_0} \left(\frac{f(z)}{g(z)} \right) = \frac{f(z_0)}{g'(z_0)}.$$

Proof. If z_0 is a simple zero of g then $g(z) = (z-z_0)\tilde{g}(z)$ where $\tilde{g}(z) \neq 0$ in a neighborhood of z_0. Then, we have

$$\frac{f}{g} = (z-z_0)^{-1} \frac{f}{\tilde{g}}.$$

Since $\frac{f}{\tilde{g}}$ is holomorphic at z_0 and $\frac{f(z_0)}{\tilde{g}(z_0)} \neq 0$, we have that $\frac{f}{g}$ has a simple pole at z_0. Then, we apply what we've seen before to get

$$\operatorname{Res}_{z_0} \left(\frac{f}{g} \right) = \lim_{z \to z_0} \frac{f(z)(z-z_0)}{g(z) - g(z_0)}.$$

We notice that we have the definition of the derivative of g at z_0 in the denominator, so we have

$$\operatorname{Res}_{z_0} \left(\frac{f}{g} \right) = \frac{f(z_0)}{g'(z_0)}.$$

□

Remark 3.22:

Suppose f has a pole of order n at z_0. Then we know in some $D_r^*(z_0)$, we have

$$f(z) = P(z) + G(z).$$

Let C be any circle which is contained in $D_r^*(z_0)$ and is positively oriented. Then

$$\int_C f(z) \mathrm{d}z = \int_C P(z) \mathrm{d}z + \int_C G(z) \mathrm{d}z.$$

By Cauchy's theorem, we have $\int_C G(z) dz = 0$. Thus, we have

$$\int_C f(z) \mathrm{d}z = \int_C P(z) \mathrm{d}z = 2\pi i a_{-1} = 2\pi i \operatorname{Res}_{z_0}(f).$$

Theorem 3.23: **Residue Formula**

Let $\Omega \subset \mathbb{C}$ open and $Z = \{z_0, \dots, z_n\}$ a finite set in Ω. Suppose $f \in \mathcal{H}(\Omega \setminus Z)$ and f has poles at $z_0, \dots, z_n$. Let γ be any circle in Ω with positive orientation such that $\gamma \cap Z = \varnothing$. Let D be the open disk bounded by γ. Then

$$\int_\gamma f(z)\mathrm{d}z = 2\pi i \sum_{z_i \in Z \cap D} \mathrm{Res}_{z_i}(f).$$

Proof. Lets first assume that f is holomorphic in Ω except for z_0 which is inside γ. As seen before, we can write $f(z) = P_{z_0}(z) + G(z)$, where $P_{z_0}(z)$ is the principal part of f at z_0 and $G \in \mathcal{H}(D_r(z_0))$. Then, we have by Cauchy's theorem (since $z_0 \notin \gamma$), that

$$\int_\gamma G(z)\mathrm{d}z = 0 = \int_\gamma f(z)\mathrm{d}z - \int_\gamma P_{z_0}(z)\mathrm{d}z.$$

Thus, we are left to prove that $\int_\gamma P_{z_0}(z)\mathrm{d}z = 2\pi i \operatorname{Res}_{z_0}(f)$. But by Cauchy integral formula, we have for $\gamma = \partial D \subset \Omega$. For $g \in \mathcal{H}(\Omega)$, we have

$$g^{(n)}(z_0) = \frac{n!}{2\pi i}\int_\gamma \frac{g(z)}{(z-z_0)^{n+1}}\mathrm{d}z.$$

Applying this to $g = 1$, we have

$$\int_\gamma \frac{1}{(z-z_0)^{n+1}}\mathrm{d}z = 2\pi i \delta_{n,0}.$$

Thus we get our desired result since $P_{z_0}(z) = \sum_{n=1}^m \frac{a_{-n}}{(z-z_0)^n}$. For the general case, let $z_1, \dots, z_n$ be the poles of f inside γ. For each z_i, let P_i be its principal part. Define

$$g(z) \coloneqq f(z) - \sum_{i=1}^n P_i(z) \quad \forall z \in \Omega \setminus Z.$$

Then $g \in \mathcal{H}(\Omega \setminus Z)$ and can be extended holomorphically to all of Ω.

To see this, fix i. Let $z_i \in F, r > 0$ such that $D_r^*(z_i) \subset D$ and $D_r^*(z_i) \cap Z = \varnothing$. Then

$$g(z) = f(z) - P_i(z) - \sum_{l \neq i} P_l(z) \quad \forall z \in D_r^*(z_i).$$

This sum is holomorphic in $D_r(z_i)$ and $f(z) - P_i(z)$ is holomorphic in $D_r^*(z_i)$.

By Cauchy's theorem $\int_\gamma g(z)\mathrm{d}z = 0$. Near each z_j, $f = P_j + G_j$ where G_j is holomorphic, including z_j. The extension is explicitly given as

$$\begin{cases} f(z) - \sum_l P_l(z) & z \in \Omega \setminus Z \\ G_j(z) - \sum_{l \neq j} P_l(z) & z = z_j \end{cases}.$$

Since $z_i \notin \gamma$, we have $\int_\gamma P_i(z)\mathrm{d}z = 0$. Thus, we have

$$\int_\gamma f(z)\mathrm{d}z = \int_\gamma g(z)\mathrm{d}z + \sum_{i=1}^n \int_\gamma P_i(z)\mathrm{d}z = 2\pi i \sum_{i=1}^n a_{-1}^{(i)} = 2\pi i \sum_{i=1}^n \mathrm{Res}_{z_i}(f).$$

□

Example 3.24:

We want to compute the integral

$$I = \int_{|z=3|} \frac{\mathrm{d}z}{(z^2+1)^2}.$$

By the residue formula, we have

$$I = 2\pi i \left(\mathrm{Res}_i(f) + \mathrm{Res}_{-i}(f)\right) = 2\pi i \left(\frac{1}{4i} - \frac{1}{4i}\right) = 0.$$

Example 3.25:

We want to compute the integral

$$I = \int_{|z|=3} \frac{z^3}{z^2+1}\mathrm{d}z.$$

By the residue formula, we have

$$I = 2\pi i \left(\frac{i^3}{2i} + \frac{(-i)^3}{-2i}\right) = 2\pi i.$$

Index

analytic continuation, 13
argument, 2

Cauchy-Riemann equations, 4
closed curve, 6
closed disc, 2
complex conjugate, 1
connected, 3

differentiable, 3
differential, 3
disconnected, 2
disk of convergence, 5
domain, 3

entire, 3
essential singularity, 18

holomorphic, 3

imaginary part, 1
isolated singularity, 17

Jacobian, 3

limit point, 12
line integral, 6
locally uniformly convergent, 15

modulus, 2

norm, 2

open disc, 2
order, 12, 18

parametrized curve, 6
piecewise smooth, 6
polar coordinates, 2
pole, 18
primitive, 7
principal argument, 2
principal part, 19

radius of convergence, 5
real part, 1
region, 3
removable, 17
reparametrization, 6
residue, 19

simple, 6
simple pole, 18
smooth, 6

toy contours, 10

uniformly convergent on compacta, 15